Predicting Chronic Disease Onset in Older Adults: A Machine Learning Study using The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Background: Chronic disease and functional decline are major health challenges for older adults, yet how well self-reported health burden can predict disease onset across longitudinal waves remains underexplored.

Methods: Using five waves of data (2009–2019), a longitudinal machine learning framework evaluated two tasks: predicting any chronic disease onset across three designs (single wave, two pooled multi-wave), and 24 disease specific outcomes. Four model families were compared via cross-validation (CV) area under the curve (AUC), with predictor contributions assessed by bootstrapped add-on analysis.

Results: Pooled multi-wave modelling improved by 22% over the single-wave baseline (0.718 ± 0.006 vs 0.589 ± 0.072), with cross-fold variance decreasing more than tenfold. Disease specific models achieved test performance up to 0.721 for functional outcomes. Reducing to 20 most important predictors caused no measurable performance loss, while cognitive, psychosocial, nutritional, and behavioural add-on variables showed no statistically significant improvement.

Conclusions: Among community dwelling older adults, chronic disease risk prediction was primarily driven by self-reported burden indicators, particularly medication use, pain, and self-rated health rather than domain specific biomarker or lifestyle profiles. These findings suggest that publicly available survey data can support meaningful risk stratification with a limited feature set.

Keywords: ageing, Machine Learning (ML), predictive modelling, chronic disease, functional decline, health burden, The Irish Longitudinal Study on Ageing (TILDA), Extreme Gradient Boosting (XGB).

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AI Declaration

I acknowledge the use of GitHub Copilot Student to assist with code completion, debugging, and LaTeX formatting during the preparation of this thesis. I also acknowledge the use of Gemini, Microsoft Copilot, Grammarly, and ChatGPT to support general brainstorming, writing preparation, literature organisation, and LaTeX formatting support.

Prompt examples are included in the auxiliary supporting material submitted alongside the final dissertation submission.

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Contents

List of Tables

List of Figures

List of Abbreviations

ADL Activities of Daily Living
AI Artificial Intelligence
AP Average Precision
AUC Area Under the ROC Curve

BMI Body Mass Index
BP Blood Pressure from Sphygmomanometer
brainPAD Brain predicted age difference

CAPI Computer Assisted Personal Interview
CES-D Center for Epidemiologic Studies Depression Scale
CI Confidence Interval
CRP C-Reactive Protein
CSV Comma Separated Values
CV Cross Validation

DM Diabetes Mellitus
DNN Deep Neural Network

EU European Union

FI Frailty Index
FOF Fear of Falling
FR Frailty

G2G Gateway to Global Ageing Data
GBM Gradient Boosting Machine
GDPR General Data Protection Regulation
GRIP Grip Strength
GSR Gait Speed Reserve

HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength
HPO	Hyperparameter Optimization
HRS	Health and Retirement Study, (a longitudinal study of ageing in the United States)
IADL	Instrumental Activities of Daily Living
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
LR	Logistic Regression
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMSE	Mini-Mental State Examination
MOCA	Montreal Cognitive Assessment
MRI	Magnetic Resonance Imaging
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PC1	First Principal Component
PCA	Principal Component Analysis
PMF	Pseudonymised Microdata File
PR-AUC	Precision Recall Area Under the Curve
PRISMA	Preferred Reporting Items for Systematic reviews and Meta Analyses
RAND	Research and Development Corporation
RF	Random Forest
ROC	Receiver Operating Characteristic
SCQ	Self Completion Questionnaire
SHAP	Shapley Additive Explanations
SHARE	Survey of Health, Aging and Retirement in Europe

SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)
UCLA	University of California Los Angeles
UGS	Usual Gait Speed
UL	University of Limerick
WHO	World Health Organization
XGB	Extreme Gradient Boosting

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [7–9]. Understanding how these multidomain factors interact over time is essential for early risk identification.

Lifestyle risk factors are estimated to contribute over $514 billion in direct and indirect healthcare costs annually across Europe [10], and in Ireland, malnutrition alone is estimated to cost over €1.4 billion per year in hospital and community care costs [11]. Frameworks such as the World Health Organization (WHO) *Global Action Plan* and Ireland's *Healthy Ireland* strategy identify modifiable health behaviours as central to disease prevention.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [12], Physical Activity (PA) [13], Scales and Questions on Mental Health and Wellbeing (MH) [14], and Social Variables (SOC) [15] improve predictive performance.

The Irish Longitudinal Study on Ageing (TILDA)[1] provides a nationally rep-

[1] TILDA's data are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to University of Limerick (UL) Science & Engineering ethics

resentative cohort of adults aged 50 and over in Ireland, surveyed repeatedly across physical, clinical, functional, psychological, and social domains. An externally produced Research and Development Corporation (RAND) harmonised TILDA dataset (waves 1–2, the only waves currently harmonised), aligned to international standards, was inspected and preprocessed alongside the five core waves.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture risk trajectories, and insufficient attention to the preprocessing and representation of heterogeneous longitudinal health data.

This thesis addresses these gaps by applying ML to pooled and cross-wave TILDA data, evaluating whether richer predictor sets, longitudinal designs, and reproducible preprocessing improve chronic disease onset prediction in older adults.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutritional factors, and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

- **Limited integration of multidomain predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [12, 16].
- **Underuse of longitudinal dynamics:** Few studies use repeated measures to track how risk changes over time, such as the progression of Frailty (FR) [6]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$.
- **Healthcare data preprocessing:** Longitudinal health survey data present significant representation challenges. Variables encoding the same concept

approval and data retention requirements.

may differ in format, value ordering, and coding scheme across waves, while health specific constructs such as frailty index items require careful ordinal handling to preserve clinical meaning during numerical conversion [17]. These challenges are rarely reported in the ML literature, yet the quality of downstream predictions depends directly on how accurately the data are prepared.

These gaps motivate the multi-wave, multidomain approach taken in this thesis, which uses TILDA's five longitudinal waves to test whether richer predictor sets and pooled designs improve chronic disease onset prediction.

1.2 Research aims and objectives

The goal of this thesis is to develop and evaluate longitudinal ML approaches that combine multidomain predictors from TILDA to predict chronic disease onset in older adults, while identifying the features and modelling designs that best support this task.

This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, deriving multidomain features across physical, functional, clinical, lifestyle, and psychosocial domains.

- Develop and compare a linear baseline (Logistic Regression (LR)) against tree ensemble and neural network families (Extreme Gradient Boosting (XGB), Random Forest (RF), Multi-Layer Perceptron (MLP)) across single wave and pooled longitudinal designs.

- Evaluate models using Area Under the ROC Curve (AUC), calibration, and classification metrics across all designs.

- Identify the most informative predictors of chronic disease onset and, through a predictor add-on analysis, assess whether domain specific features contribute independent predictive value beyond the core health burden set.

- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates that longitudinal pooling substantially improves predictive performance and stability for chronic disease onset prediction in this cohort, and identifies the compact set of health burden indicators that drive this signal. Nutritional, psychosocial, and behavioural domain features added no independent predictive value beyond this core set, a finding discussed in Chapter 5.

- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating features into ML workflows. Full pipeline implementation, including the cleaning engine, scenario detection scripts, and modelling notebooks, is available at `https://gitlab.com/abderahman-msc-project/milestones/-/blob/main/README.md` (private repository, access granted on request).

- **Clinical:** Identifies the most informative predictors of chronic disease onset and evaluates the marginal contribution of domain specific features, providing a data driven basis for risk stratification.

1.4 Thesis outline

Chapter 2 reviews ML approaches to chronic disease prediction in ageing populations, with emphasis on lifestyle, nutrition, and methodological considerations.

Chapter 3 describes the data preparation process, including the TILDA study design, data collection, thematic domain structure, and the manual variable screening workflow. In *Chapter* 4, the automated cleaning pipeline is presented, including the variable scenario detection system, the deterministic six step cleaning sequence, and validation of the pipeline against published benchmarks. *Chapter 5* presents the empirical evaluation, covering performance across three modelling designs, disease specific prediction, predictor add-on analysis, Shapley Additive Explanations (SHAP) based interpretability analysis, and an interpretation of the health burden correlation structure underlying the results. *Chapter 6* discusses the implications and future directions of the research.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It covers four interrelated domains: physical function and FR, MH, nutrition and biological ageing, and ML applications for chronic disease prediction.

2.1 Search strategy and study selection

This review applied systematic mapping principles [1]: research questions were defined, relevant databases searched, inclusion and exclusion criteria applied, and findings synthesised through an integrative approach [2, 18]. The primary focus was chronic disease prediction in older adults using TILDA. Where ML applications specific to TILDA were limited, the scope was extended to include comparable ageing cohort studies.

2.1.1 Database search and query development

The main topic of interest was chronic disease onset prediction in older adults using ML, with TILDA as the primary data source. Relevant databases (e.g., ScienceDirect and IEEE Xplore) were searched using combinations of keywords such as "Irish Longitudinal Study on Ageing", "machine learning", "chronic disease", "nutrition", "diet", and "physical activity", as shown in Table 2.1.

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

Table 2.1: Database searches and number of studies per database [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing) AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	9

Some search strings returned few or irrelevant entries, particularly for ML applications in TILDA, highlighting a key research gap. Iterations continued until suitable combinations consistently retrieved relevant studies. Duplicates were removed and the remaining studies screened individually for relevance to ageing, chronic disease, and ML. Key elements for each study, including model type, features used, and main findings, were recorded in Table 2.3. The full selection process is illustrated in the Preferred Reporting Items for Systematic reviews and Meta Analyses (PRISMA) diagram (Figure 2.2).

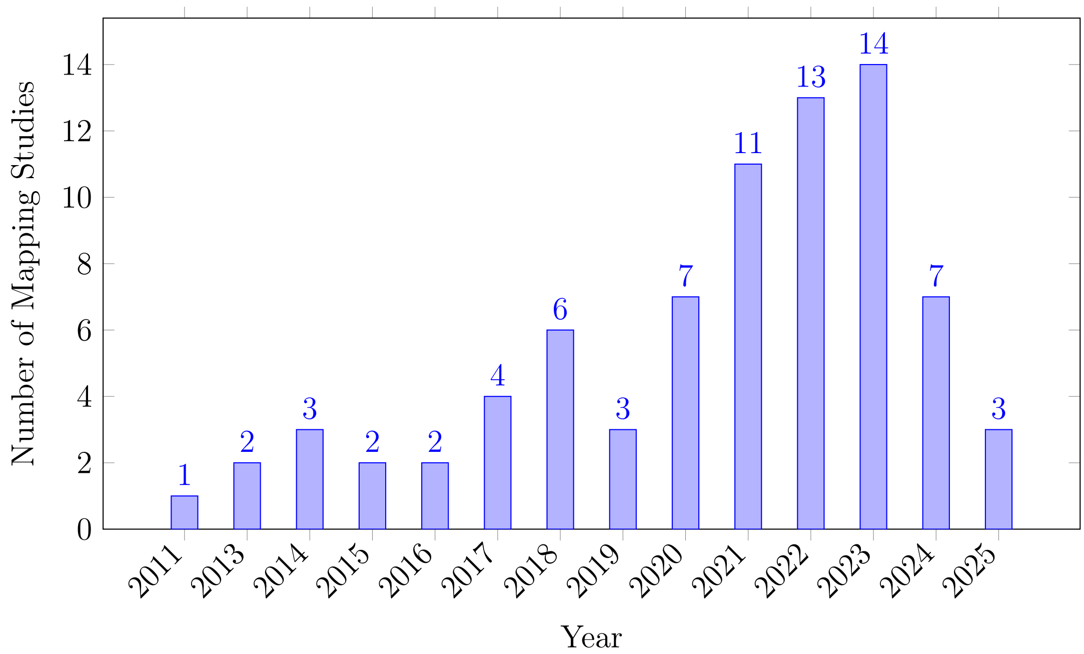

Figure 2.1: Publications per year.

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customised version of the Newcastle-Ottawa Scale (NOS) was applied (Table 2.2). The NOS evaluates studies across three domains: *selection* (S1–S4), *comparability* (C1–C2), and *outcome* (O1–O3). Selection criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. Comparability examines the control of key confounders (variables that may influence both the predictor and outcome), while outcome evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored binary (0 = criterion not met, 1 = criterion met), and the total reflects overall study quality.

This initial table guided refinement into two subsets: studies critically related to TILDA, and studies applying ML within TILDA for predictive modelling. The

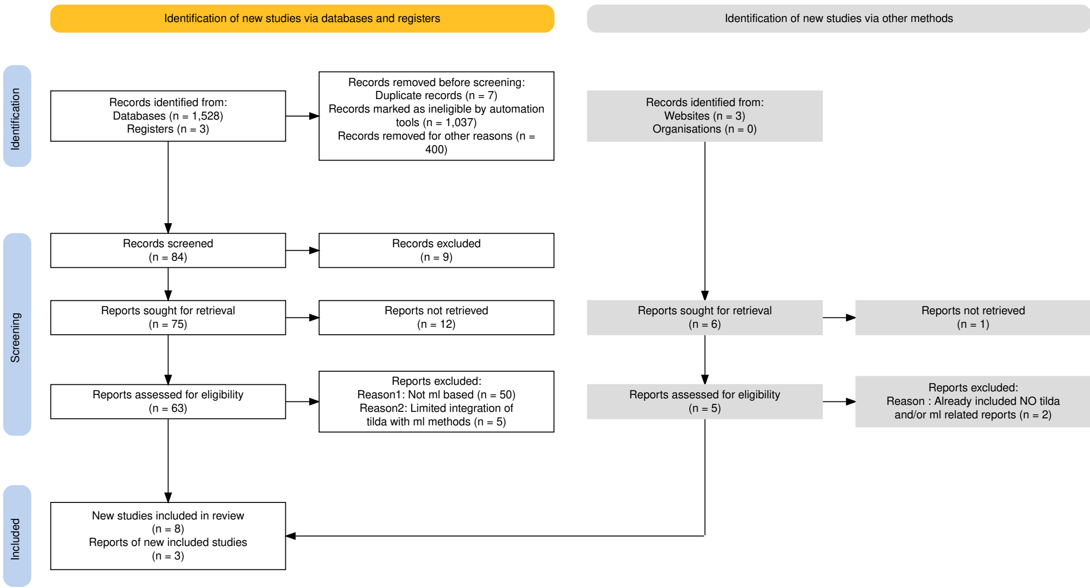

Figure 2.2: PRISMA chart for systematic reviews showing the study selection process for the systematic mapping and integrative review [5].

second subset revealed minimal integration between ML methods and TILDA data.

Data were extracted and synthesised to identify methodological trends, including Markov models, regression baselines, and Cross Validation (CV) protocols [2]. Given the limited number of ML studies using TILDA directly, the review was broadened to include chronic disease prediction research in comparable older adult populations.

2.2 Physical function and frailty

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality [24].

At the same time, FR indicators are highly correlated with one another. When grip strength declines, gait speed and functional capacity typically follow, mean-

Table 2.2: Quality evaluation of included systematic literature reviews [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [19])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [20])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [6])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [7])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [21])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [22])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [23])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [24])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [25])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [26])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [27])	1	1	1	1	0	0	1	1	1	7

ing each additional predictor captures diminishing independent information. This correlation structure has direct implications for ML models, which must decide how many such indicators to include before returns become negligible, a question revisited empirically in Chapter 5.

2.2.1 Probabilistic interpretation of frailty

From a probabilistic perspective, the likelihood of an adverse outcome Y (such as mortality) given observed FR features X is expressed as a conditional probability $P(Y \mid X)$, and identifying which features carry the strongest predictive signal requires understanding how much unique information each contributes.

When features X_i and X_j are statistically independent,

$$P(X_i, X_j) = P(X_i)\, P(X_j),$$

and each adds fully independent information to the model. In practice, however, FR related predictors such as UGS, grip strength, and mobility limitation are often highly correlated, so the features share information and the additional gain from including a second correlated predictor is reduced. This has direct implications for feature selection in ML, as discussed in Chapter 5.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail,

Table 2.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
Frailty (FR)	Romero-Ortuno (2021) [24]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion (EX), Unexplained weight loss (WL), Weakness (WK), Slowness (SL), Low physical activity (LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + Machine Learning (ML)	Donoghue (2023) [20]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	AUC values were ≤ 0.69 and Precision Recall Area Under the Curve (PR-AUC) ≤ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [28]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with Fear of Falling (FOF) had no recent falls in the last 12 months	Self-reported; cross sectional
Functional Mobility	Huang (2019) [29]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [30]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective LR	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self-reported falls; short follow up; mediation effect partially explored
	Laird (2023) [16]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self-reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [12]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal LR	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [25]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	LR; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational (short follow up (2 years)); limited generalisability >65 years, self-reported
ML	Xu (2023) [19]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, Extereme randomised trees (XRT), General linear model (GLM), Gradient Boosting Machine (GBM), Deep Neural Network (DNN); SHAP explainability	AUC=0.854 for 2 year Diabetes Mellitus (DM) prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [7]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [26]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised LR; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [21]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

although established FR strongly predicts mortality [24]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [31]. The presence of metabolic syndrome raises $P(\mathrm{FR} \mid \text{Metabolic Syndrome})$ and is associated with greater risk of adverse outcomes over time [31]. Age also plays an important role: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [29].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity motivates probabilistic models that can update beliefs about risk as new observations arrive, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X. In practice, causal inference remains limited by observational designs and predictors are frequently examined in isolation. Although formal Bayesian inference is not implemented in this study, this probabilistic framing motivates the use of well calibrated ML models evaluated via AUC and the Brier score, as described in Chapter 5.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. Participation in physical, SOC, and cognitive activities, summarised by the Act-Belong-Commit (ABC) indicator, was associated with lower depression and anxiety, suggesting that SOC integration and lifestyle protect mental wellbeing [32]. However, self-reported measures may overestimate such associations, raising concerns about response bias. These two findings conflict: the effect of SOC engagement is strong when self-reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [13]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, mediated by FOF [30]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing requires longitudinal, multidimensional approaches capable of capturing bidirectional pathways. Formally, the conditional probability

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement})$$

quantifies how these interacting features jointly influence MH outcomes, indicating which predictors carry the strongest signal and which may be redundant.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers such as vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. Higher plasma lutein and zeaxanthin, dietary carotenoids found in leafy vegetables, were associated with slower FR progression [12], while vitamin D deficiency predicted DM and musculoskeletal decline, particularly among adults with low sunlight exposure [33]. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation. Whether plasma biomarkers independently improve prediction beyond self-reported health indicators in a general community sample is the kind of question that motivates the add-on analysis in Chapter 5.

2.4.1 Contextual and broader nutritional considerations

Malnutrition also reflects SOC and functional risks. Hospitalisation, functional loss, and poor SOC support were found to increase malnutrition risk, with notable sex differences [25], highlighting that biological markers are often studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy links metabolic syndrome and nutritional deficits to accelerated biological ageing, as indexed by epigenetic clocks [34]. Although most of this evidence is cross sectional and cannot capture change over time, these indicators offer a more precise measure of vulnerability. Reliability is also reduced in biomarker studies with small sample sizes, limiting confidence in the estimated effect magnitudes.

Nutrition interacts with biological, metabolic, and SOC systems simultaneously. Studying one nutrient at a time risks missing this bigger picture. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by ML pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine learning for chronic disease prediction

ML is increasingly applied in ageing research because it can capture nonlinear and multidimensional relationships that traditional regression models, such as LR, often miss [19, 21]. Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables.

A recurring pattern is that models rely primarily on readily measured clinical data such as comorbidity counts, mobility scores, and vital signs, while nutrition, SOC, and lifestyle predictors remain underrepresented [6, 19]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially given that MH and nutrition exert independent and interacting effects on later life outcomes.

2.5.1 Methodological approaches

The methodological approaches used in ageing research span both traditional and advanced models. LR and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture nonlinearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, GBM, as well as DNNs, which generally achieve higher predictive accuracy.

For instance, LR for mortality prediction in TILDA achieved an AUC of 0.78 [35], whereas ensemble methods for DM incidence reached 0.854 [19].

2.5.2 Performance and validation

Predictive performance across studies spans a wide range, with AUC values between 0.64 and 0.85 for outcomes including MH, diabetes, falls, and mortality. This spread raises a question worth asking: Why does performance vary so much between what appear to be similar tasks? One underappreciated reason is how health data are preprocessed before modelling. Longitudinal survey variables often contain mixed formats, inconsistent coding, and wave specific encodings that, if handled incorrectly, can silently distort model inputs. Studies rarely report preprocessing decisions in enough detail to assess whether performance differences reflect genuine signal or encoding artefacts. This reproducibility gap is a methodological limitation that this thesis directly addresses through the deterministic cleaning pipeline described in Chapter 4.

Furthermore, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues reduce the generalisability of existing models [19].

2.5.3 Explainability in machine learning for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent across studies, particularly in deep learning models [4, 7, 19].

These methods improve transparency by revealing which features drive predictions, yet black box concerns persist, particularly in deep learning, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice.

2.6 Gaps in the literature

Five gaps emerge from the reviewed literature that this thesis directly addresses.

First, existing studies applying ML to TILDA have largely used single waves or short follow-up windows, limiting the ability to capture how risk changes across time [6, 20]. This thesis uses all five available waves, enabling pooled longitudinal designs that test whether temporal structure improves prediction. Cross study replication using the RAND harmonised TILDA dataset, which maps variables to international standards compatible with cohorts such as Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS) and Survey of Health, Aging and Retirement in Europe (SHARE), is identified as a direction for future work.

Second, predictive models in ageing research overwhelmingly rely on physical health indicators such as comorbidity counts and mobility measures, while nutrition, SOC, and behavioural variables are rarely integrated. As shown by [12] and [25], nutritional and social factors independently influence health trajectories. Yet it remains an open empirical question whether these domains add predictive value beyond established clinical indicators in a general community sample. This thesis assesses that question directly through a predictor add-on analysis in Chapter 5.

Third, the preprocessing and numerical representation of longitudinal health survey data is a genuinely difficult problem that receives little attention in the ML literature. Variables encoding the same concept may differ in format, value ordering, and coding scheme across waves. Health specific constructs such as Frailty Index (FI) items require ordinal aware handling to preserve clinical meaning, and incorrect encoding can silently distort downstream predictions. Few published studies describe their preprocessing workflow in sufficient detail to enable reproducibility or cross study comparison. This challenge is addressed through a transparent deterministic six step cleaning pipeline applied consistently across all TILDA waves, with full audit documentation described in Chapter 4.

Fourth, many outcomes relevant to older adults, including chronic pain and subjective wellbeing, remain underexplored as prediction targets. Standard focus on clinical diagnoses omits outcomes that are often more meaningful for intervention planning and quality of life. This thesis includes pain, functional decline, and hearing loss alongside ICD-10 coded disease outcomes described in Section 5.1.1.

Finally, where longitudinal designs have been used, cross wave validation is almost always internal. Weak validation and narrow feature sets limit the generalisability of existing models and increase the risk of overfitting. This thesis applies participant level train/test splitting with stratified group CV to prevent data leakage across all designs.

2.7 Chapter summary

The evidence reviewed in this chapter shows that FR, mental health, and nutrition do not operate in isolation. FR is partly reversible but remains strongly associated with mortality. MH both influences and is shaped by physical decline and nutrition interacts with biological, metabolic, and social systems in ways that single variable studies frequently underestimate.

While ML has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and achieving robust longitudinal validation. An often overlooked challenge is the preprocessing of health survey data itself, where inconsistent formats and wave specific encodings can compromise model inputs if not handled carefully.

These gaps motivated the integrative, longitudinal approach developed in Chapters 3, 4, and 5.

Chapter 3

Methodology

3.1 Study population and design

TILDA provides longitudinal data across waves for community dwelling adults in Ireland. The RAND harmonised TILDA dataset (waves 1–2), aligned to cross study standards via the Gateway to Global Ageing Data (G2G) framework, was also cleaned using the same pipeline.

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for ML, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering occurred primarily at the *variable level* during the manual screening stage, which excluded features with extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

This approach prioritises data integrity and sample size over the strict exclusion criteria common in traditional epidemiological studies.

3.2 TILDA data collection

TILDA data collection occurred over five longitudinal waves (Table 3.1).

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Computer assisted personal interviews conducted in participants homes, covering health status, sociodemographics, economic circumstances, and family networks. This assisted format ensures automatic routing between questions, eliminates data entry errors, and allows real time consistency checks, improving data integrity compared with paper based methods.

2. **Self Completion Questionnaire (SCQ):** Paper based questionnaires completed privately by participants, capturing sensitive information including mental health, alcohol use, sexual activity, and subjective wellbeing. The self completion format reduces social desirability bias for topics participants may be reluctant to discuss with an interviewer.

3. **Health Assessment:** Objective clinical and biomarker measurements, including blood pressure, grip strength, gait speed, cognitive tests, and blood samples, conducted in dedicated health assessment centres (waves 1 and 3) or adapted for home measurement in subsequent waves.

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

3.3 Thematic domains

Variables were organised into seven thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the Pseudonymised Microdata File (PMF) metadata (Table 3.2), indicates its source or research domain.

Table 3.1: Dimensions of raw TILDA and harmonised datasets version 5.5 (prior to cleaning in Section 4.1).

Dataset	Collection Period	Participants (N)	Raw Variables (N)
Wave 1 [36]	2009–2011	8501	1619
Wave 2 [37]	2012–2013	7206	724
Wave 3 [38]	2014–2015	6397	1029
Wave 4 [39]	2016	5713	990
Wave 5 [40]	2018–2019	4978	983
Harmonised [41]	2016 (release)	8501	572

This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and ML.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code. Variables with multiple responses or loops include an underscore and number to specify the response or iteration.

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area. These prefixes provide a consistent framework for categorising features and guiding the research into thematic domains for downstream analysis. Additional derived variables, constructed from CAPI or SCQ responses, are documented in a separate codebook that specifies their source items and scoring logic. These derived variables were assigned thematic domains based on the research area of their component items.

By categorising the raw variables into these seven thematic domains, a structured foundation was established for the technical dimensionality reduction and cleaning pipeline described in the following Section 3.4 and Chapter 4.

3.4 Manual variable screening

Prior to any data cleaning or modelling, a structured multi-stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets.

Table 3.2: PMF v5.5 derived variable prefixes [3].

Variable Prefix	Research Area
1: Socio Demographic & Context	
INC	Income variables
SES	Social class
SOC	Social
2: Physical Health & Clinical Baseline	
BP	Blood pressure from sphygmomanometer
MD	Medication data
SCR	Screening tests
3: Functional Status & Frailty	
ADL	Activities of daily living
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
GRIP	Grip strength
IADL	Instrumental activities of daily living
4: Lifestyle & Behavioural	
BEH	Health behaviours
5: Nutrition & Metabolic Biomarkers	
BLOODS	Results of blood sample analysis
6: Psychological & Cognitive	
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
MH	Scales and questions on mental health and wellbeing
Other / Administrative	
G2G	Gateway to Global Aging Data

As shown in Table 3.3 (for example, 1619 $\rightarrow$ 809 variables in wave 1), the raw PMF files were reduced to a subset of variables of interest. This procedure also separated automated diagnostic profiling from manual variable exclusion, ensuring transparency in all selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Due to compatibility issues, the harmonised dataset was loaded from a (`Stata`) (`.dta`) file, while waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (`.sav`) files.

2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds described below. Variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.

4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained *variables of interest*.

5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No scripts were used to clean, recode, or impute data at this stage.

Manual filtering was applied according to the following criteria (Figure 3.1):

- **Missingness:** Variables with $> 80\%$ missing values were dropped. This threshold retains sufficient feature coverage while avoiding the imputation uncertainty that arises when the majority of values are absent.

- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.

- **Dominant category (top value/frequency):** Variables where a single category (sample value) marked as the `topValue` occurs over $\approx > 90\%$ (`topFrequency`) of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.

- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave's PMF file into Comma Separated Values (CSV) files (Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

During this process, variables exceeding these exclusion thresholds were visually flagged (e.g., marked in red). Non-flagged variables were concatenated using a spreadsheet formula that wrapped each variable name in double quotes and separated them with commas,[1] producing the correctly formatted input for the wave specific JSON files. These files represent the *variables of interest* retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing and cleaning in Section 4.1.

As shown in Table 3.3, this process yielded a significantly reduced dataset for each wave. For instance, in wave 1, manual filtering reduced the feature space by 50.03%. When considering all five waves collectively, the total number of variables was reduced from $5,345 \rightarrow 3,099$ representing $2,246$ dropped features with an overall reduction average of $\approx 42.0\%$. All variable exclusion decisions were

[1] The formula used was `=TEXTJOIN(", ", TRUE, CHAR(34) & J2:Jn & CHAR(34))`, applied in Microsoft Excel to each wave's variable list.

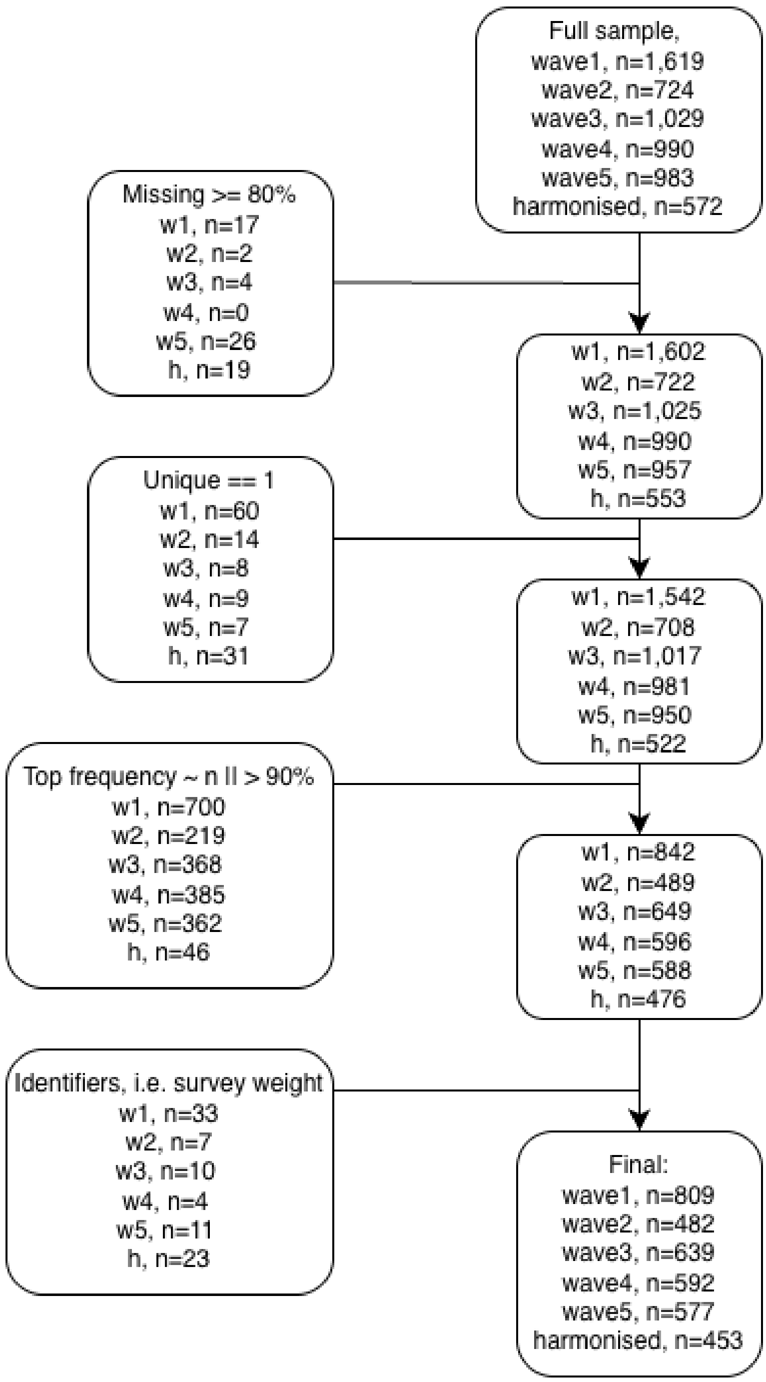

Figure 3.1: Flowchart of the manual five step workflow for variable screening and filtering. Placeholder (n) values represent the number of variables at each stage, illustrating the reduction from the raw dataset to the filtered dataset prior to automated preprocessing.

completed at this stage. Subsequent steps focus on numerical standardisation and model preparation.

3.5 Dataset characteristics after filtering

To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`.ipynb`) file implemented with `pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves. The purpose of this profiling stage was not transformation but to expose structural inconsistencies and latent formatting issues that would undermine direct numerical conversion.

Table 3.3: Input shape for cleaning pipeline (post manual filtering workflow).

Dataset	Row (N participants)	Column (N features)	% Reduction
Wave 1	8501	809	50.0%
Wave 2	7206	482	33.4%
Wave 3	6397	639	37.9%
Wave 4	5713	592	40.2%
Wave 5	4978	577	41.3%
Harmonised	8501	453	20.8%

For example, in wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns. Age distributions confirmed that the study population was restricted to older adults across waves, however, ages were bottom coded for participants younger than 50 years and top coded for those aged 65–85 years and above. Missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

These insights exposed structural inconsistencies: for instance, variables that appeared numeric in fact encoded labels such as “1.Rarely or none of the time”, and many columns contained sentinel codes (e.g., −98, _99). These patterns directly informed the design of the deterministic cleaning pipeline described in Chapter 4.

3.6 Variable coverage by thematic domain

Using the PMF prefixes described in Table 3.2 as a guide, variables were assigned to seven thematic groups in Table 3.4, each capturing a conceptually coherent domain spanning physical, clinical, cognitive, social, and behavioural aspects of ageing. The seven thematic groups integrate information from CAPI, SCQ, and derived variables, covering sociodemographic context, physical health, functional status, lifestyle behaviours, nutrition and metabolic biomarkers, psychological and cognitive measures, and administrative identifiers. The heatmap at the base of Table 3.4 shows domain coverage across waves, darker shading indicates a higher variable count highlighting which domains contribute most features to the dataset.

Table 3.4: Thematic domain grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variable Examples	Total Variables
1	**Socio-Demographic & Context**	Baseline demographic and socioeconomic indicators	Age/Sex	844
			Education	
			Income	
			Marital status	
2	**Physical Health & Clinical**	Clinical and self reported health measures	Blood pressure	839
			BMI	
			Chronic disease status	
			Self-rated health	
3	**Functional Status & Frailty**	Physical functioning and frailty markers	Activities of Daily Living (ADL)/Instrumental Activities of Daily Living (IADL)	174
			Grip strength	
			Gait speed	
4	**Lifestyle & Behavioural**	Health behaviours and activity patterns	Physical activity	120
			Smoking	

ID	Thematic Group	Description	Representative Variable Examples	Total Variables
			Alcohol use	
5	**Nutrition & Metabolic**	Metabolic and biomarker indicators	C-Reactive Protein (CRP) Creatinine Lipid markers	22
6	**Psychological & Cognitive**	Cognitive performance and mental health	Mini-Mental State Examination (MMSE) Depressive symptoms Stress scale	291
7	**Other / Administrative**	Survey metadata and structural variables	Sampling weight Completion status Identifiers (ID)	809

	SocioDemographic_Context	PhysicalHealth_Clinical	FunctionalStatus_Frailty	Lifestyle_Behavioural	Nutrition_Metabolic	Psychological_Cognitive	Other_Administrative	Total_Variables
wave1	272	149	48	58	10	97	175	809
wave2	119	148	18	21	0	38	138	482
wave3	164	184	44	14	8	93	132	639
wave4	161	186	30	13	4	36	162	592
wave5	128	186	34	14	0	27	202	591
Total_Per_Domain	844	853	174	120	22	291	809	3113

Thematic domain dominance across waves (higher n features per domain).

Chapter 4

Technical Implementation

4.1 Automated cleaning engine

The core challenge in converting heterogeneous variable types across the six datasets (Table 3.3) was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained features using a single numerical representation (`float64`) with a unified missing or non response sentinel (`-1.0`).

Categorical variables were dominant (2,916 columns) while numeric types were fragmented, with `float64` (462 columns) adopted as the common standard to support decimal values from midpoints, ordinal scaling, and frequency normalisation. The solution was implemented as two stages: automated *variable scenario detection* function (`detect_column_scenario_mapping`, (Appendix C)) and a deterministic *six step numerical cleaning engine* (Appendix E). Nine scenario categories were defined initially and three served only as validation fallbacks and were eliminated as coverage improved, leaving six active transformation steps applied uniformly across all waves (Appendix B).

4.1.1 Stage 1: Variable scenario mapping

All variables passed sequentially through the six step pipeline, most were resolved in Steps 1–5, with step 6 (Appendix D) handling only residual variables that could not be resolved by earlier transformations.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of variable states before transformation and feature representations after, including all representative sample values for each variable. This allowed direct inspection of how the same variable was represented across waves prior to cleaning, particularly in cases where identical semantic categories appeared in different orders or formats. By comparing these grouped samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the `show_grouped_vars_for_scenarios` function (Appendix F) was formatted to create an automatic friendly ".md" file for rapid, surface level verification, which honed the final mapping dictionary (an example can be found in Appendix G) for all features across all waves. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.1.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed sequentially by the *six step cleaning engine*, resolving simple and unambiguous cases first to maximise the proportion of successfully transformed variables before tackling more complex structures, particularly censored and multi-class scenarios (Figure 4.1). With later steps progressively shifting from cleaning into semantic feature engineering, this deterministic sequence applied identical transformation and feature encoding logic to each wave, with later steps (particularly Steps 4–6) performing rule based feature engineering by deriving semantically meaningful numerical representations directly from observed category tokens at runtime. The rationale for this embedded feature engineering approach is detailed in Section 4.1.3, implementation is provided in Appendix D. This design reflects the empirical diversity of survey encodings: a small subset of variables only reveals its structure once simpler transformations have been applied.

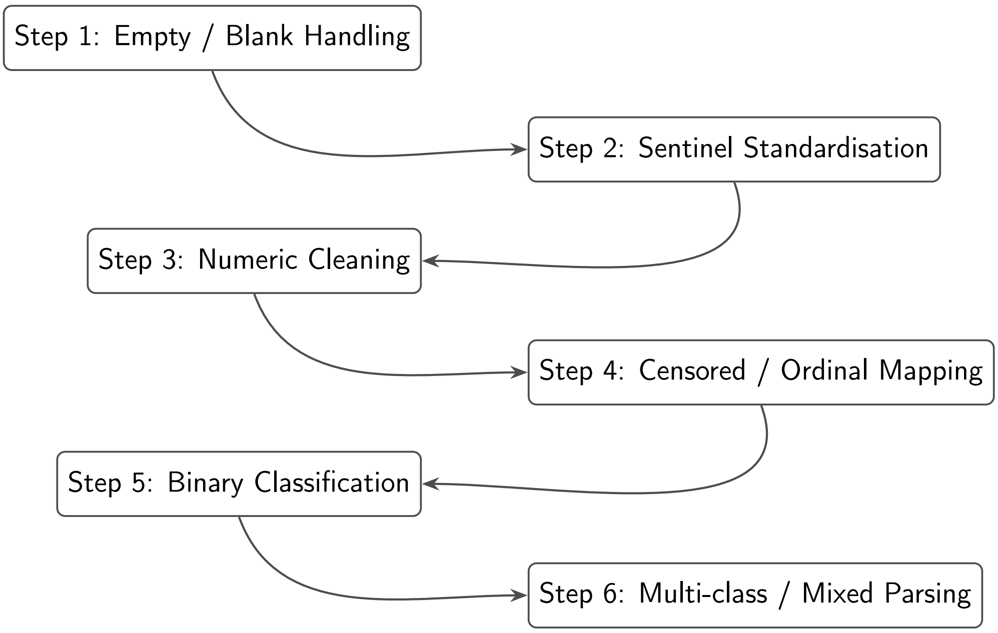

Figure 4.1: Flowchart of the final six step cleaning pipeline applied to all TILDA waves. Detailed transformation tables are provided in Appendix B.7. The corresponding implementation "`.ipynb`" script is documented in Appendix E at each scenario step.

The final cleaning pipeline consisted of six deterministic steps applied uniformly across all TILDA waves, with increasingly complex transformations handled in later stages. Steps 4–6 required particular attention due to hybrid, ordinal, or multi-class formats. A summary of each step and examples is provided below.

1. **Empty / Blank Values:** All missing entries, empty strings, and whitespace were converted to the standard non-response sentinel (`-1.0`).

2. **Sentinel Values:** Wave specific non-response codes (e.g., `-98`, `-99`) and textual sentinels (e.g., `''dk''`, `''not applicable''`) were mapped to the unified (`-1.0`) placeholder.

3. **Numeric Transformation:** Numeric like columns containing embedded units, symbols, or hybrid strings (e.g., `50+`, `<10`, `1-2.5`) were parsed using regex and converted to `float64`. For ranges, midpoints &/or boundary values were used. For censored or annotated values, appropriate upper/lower bounds were applied.

4. **Censored / Ordinal Variables:** This step addressed variables representing ordinal scales, bracketed numeric ranges, or frequency expressions (e.g., "2–3 days a week", "daily"). Transformations included:

    - Converting textual ordinal labels and Likert scale responses to numeric equivalents.
    - Mapping calendar based frequency descriptors to numeric weekly equivalents.
    - Manual verification was conducted for low count or unusual variables using the Markdown audit outputs to confirm semantic meaning was preserved.

5. **Binary Categorical Variables:** Two level features (e.g., Yes/No, Male/Female) were dynamically encoded using mappings derived from the observed value set of each variable:

    - Semantic rules were applied e.g., ("no", "not", "female" $\rightarrow$ 0.0) and ("yes", "male" $\rightarrow$ 1.0).

- Count based detection verified that variables had exactly two non-missing unique values.
- Features with more than two sample values were also analysed for sentinel presence:
  - If a variable had three observed values and one was a sentinel, it was treated as binary.
  - If a variable had four observed values and two were sentinels, it was treated as binary.
- Sentinels were preserved as `-1.0`.
- This approach ensured order invariant encoding across waves, even when original labels or numeric prefixes differed.
- *Example (sentinel case):* Feature `X` with pre-transformed values `['NOT Getting in or out of bed', -1.0, 'Getting in or out of bed']` was encoded as `[0.0, -1.0, 1.0]`, demonstrating how a sentinel preserves binary semantics.

6. **Multi-class / Mixed Variables (rule based feature engineering):** Variables with remaining multi-class structure, mixed types, or unresolved string numeric hybrids after earlier transformations were processed in the final step.

   Technical implementation for this step are provided in Appendix D:

   - Text based parsing extracted numbers, ranges, units, percentages, and frequency descriptors into numeric values, yielding interpretable numerical feature representations.
   - Sentinel handling ensured all known non-response codes mapped to (`-1.0`).
   - Original values were grouped and assigned numeric identifiers in first seen order, non numeric categorical labels were assigned integer codes above the highest observed numeric value to prevent collisions.
   - Manual inspection using the Markdown audit file verified that semantic equivalence was maintained across waves.

- Step 6 was designed to be *order invariant*, dynamically interpreting category values within each wave rather than relying on predefined lookup tables shared across waves. This preserved consistent numeric representation regardless of variable ordering or wave specific encodings.
    - *Example:* Step 5 handled simple binary variables, e.g., (any pain reported) with pre transformed values `['pain', 'No pain']` and numeric mapping `[1.0, 0.0]`. In contrast, step 6 dynamically processed multi-class variables, e.g., (pre tax wage/salary) many of which contained over 20 distinct pre transformed categories (e.g., `-1.0, '200-299', 'Less than 100', ..., '6,000-6,999'`) into consistent numeric representations (e.g., `-1.0, 249.5, 99, ..., 6499.5`). This illustrates why step 6 required wave specific, dynamic parsing to maintain semantic equivalence across diverse observed values.

4.1.3 Embedded feature engineering

Although the six step pipeline is framed primarily as a numerical cleaning procedure, steps 4–6 also perform explicit rule based feature engineering by interpreting heterogeneous categorical survey responses and converting them into semantically meaningful numerical representations. Rather than acting as a separate post processing stage, feature engineering is embedded directly within the cleaning logic, formalising ordinal structure, frequency, and magnitude encoded in text into directly modelable `float64` values suitable for downstream analysis.

Crucially, this process engineers the representation of each feature rather than expanding the feature space. Free text frequency responses such as "`once or twice a week`", "`a couple of times per month`", or "`almost every day`" are not treated as nominal categories. Instead, lexical cues (e.g., `once`, `twice`, `couple`) are resolved to numeric quantities and normalised onto a common temporal scale, yielding a continuous rate representation. Implicit semantic information such as ordering, intensity, and frequency is therefore preserved through deterministic parsing and normalisation without introducing additional variables.

This design maintains the original dimensionality of the dataset while ensuring numerical homogeneity and semantic integrity, producing a compact and model ready representation of complex real world survey data.

4.1.4 Cross wave semantic verification

Given the absence of guaranteed category ordering across waves, preserving *semantic equivalence* during numerical encoding and feature representation was essential to avoid introducing distortions into downstream ML modelling. Pre and post transformation values for each group were exported to a readable Markdown (`.md`) file. This representation confirmed that features of known importance for downstream predictive models were preserved and correctly encoded. Self-rated general health, comparative self-rated health, and smoking frequency are among the most influential predictors in FI based mortality models [6]. Retaining these features from the manual filtering thresholds applied in Section 3.4 and encoding them consistently across waves ensured that the cleaned dataset maintains coverage and semantic integrity, preventing distortions that could potentially compromise model performance.

A key challenge in cleaning the five TILDA waves was the lack of guaranteed ordering for categorical values of the same variable across different waves. Table 4.1 illustrates this for waves 4 and 5.

Gender (`sex`) variables appeared under different value encodings, with varied category orderings and numeric prefixes across waves. Values such as `['Male', 'Female']`, `['Female', 'Male']`, and prefixed forms like `["1.male', '2.female']` were all mapped to a consistent numeric representation (`1.0` for male, `0.0` for female). This ensured that semantic equivalence, rather than positional ordering or wave specific encoding, determined numerical assignment.

The Markdown audit confirmed that semantically identical categories were correctly grouped across waves regardless of variable name or encoding order, and that complex value spaces (e.g., income bands, frequency descriptors) were being collapsed to stable numeric representations consistently across waves. Full audit examples are provided in Appendix G.

Table 4.1: Inconsistent category ordering for comparative self-rated health ("How does your health compare to others your age?") across waves and its impact on positional label encoding.

Pos	Wave 4		Wave 5	
	Category	Code	Category	Code
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

4.2 Validation of the transformed features

Validation assessed whether the pipeline preserved meaningful biological structure by checking whether known relationships reported in the literature could be recovered from the cleaned data.

Due to restricted access to certain outcome variables within TILDA, full end-to-end replication of all published models was not always feasible. In particular, access through the hotdesk facility for the official mortality outcomes is subject to additional governance requirements, including institutional GDPR training certification and approval by the TILDA data access committee [42].

As a result, validation focused on two complementary strategies:

(i) Internal consistency checks against established biological and clinical relationships, and

(ii) Constrained replication of published modelling frameworks using available predictors and the derived *8-year mortality proxy* outcome.

Together, these validation steps demonstrate that the cleaning pipeline retains expected relational patterns and produces datasets appropriate for the downstream modelling tasks presented in Chapter 5.

4.2.1 Internal feature relationships

The first validation stage assessed whether cleaned predictors preserved biologically and clinically plausible relationships reported in prior TILDA studies. This analysis focused on predictors identified by [4] as accounting for approximately 95% of the explained variance in Gait Speed Reserve (GSR), defined as the difference between maximum and UGS.

As GSR is not directly available in the public wave 3 dataset, validation was performed by examining the directional relationships between the 7 primary predictors themselves. This approach evaluates whether the encoding process, including sentinel handling, missing value treatment, and ordinal mapping, preserved the expected structure of the data rather than simply fitting to a particular outcome.

Figure 4.2 presents the pairwise Pearson correlation matrix for the selected predictors after the deterministic six step cleaning pipeline. Grip strength exhibited a negative association with both age and chair stand time, reflecting the inverse relationship between muscular strength and physical performance decline. Cognitive performance (Montreal Cognitive Assessment (MOCA)) demonstrated a moderate negative correlation with age and a positive association with educational level, consistent with known patterns of cognitive reserve in older adults. Education correlated positively with cognitive performance, and BMI showed the expected weak negative association with age, reflecting reduced physiological reserve in older cohorts.

Table 4.2 summarises the expected and observed directional relationships between key predictor pairs. Notably, grip strength showed moderate negative correlations with both age ($r = -0.265$) and chair stand time ($r = -0.170$), reflecting the expected inverse relationship between muscular strength and physical performance decline. Cognitive performance (MOCA) displayed a moderate negative association with age ($r = -0.381$) and a positive association with education ($r = 0.367$), consistent with known patterns of cognitive reserve in older adults. BMI demonstrated the anticipated weak negative correlation with age ($r = -0.073$). These results confirm that the cleaning pipeline retained the expected numerical structure and relational patterns of the predictors, rather than introducing artificial associations.

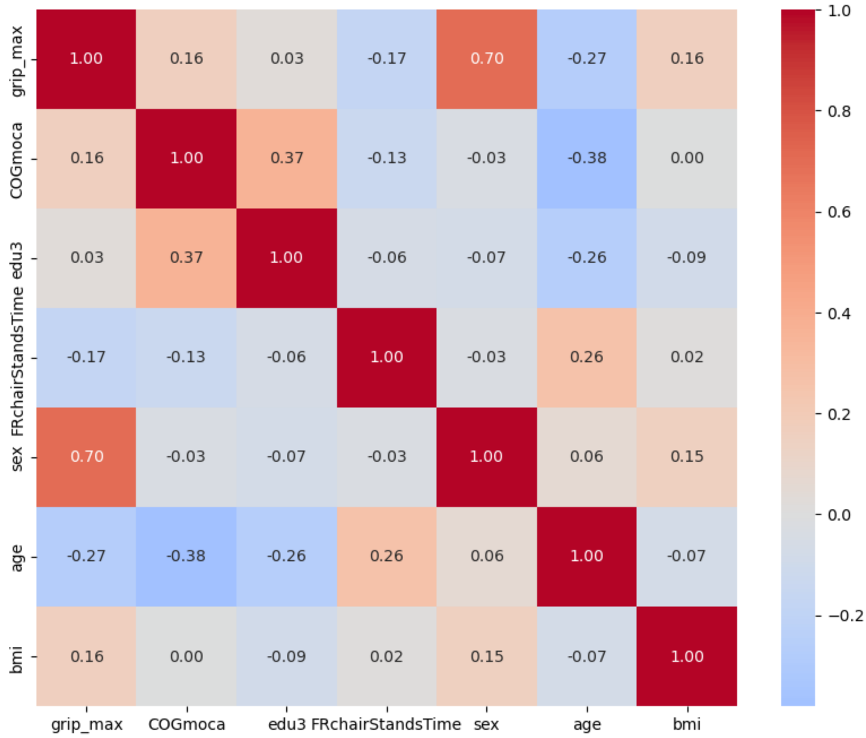

Figure 4.2: Pairwise Pearson correlations for the seven predictors used in directional validation (wave 3).

4.2.2 Model based validation

To evaluate whether the cleaned dataset preserves predictive utility, a constrained replication of [6] was conducted using available wave 1 predictors. The study predicted 8-year mortality using 32 FI items. In the cleaned wave 1 dataset following the deterministic six step cleaning pipeline described in Section 4.1.2, only 16 of these 32 FI items remained following the manual variable filtering described in Section 3.4, which prioritised global numeric consistency across waves.

An *8-year mortality proxy* outcome was constructed to approximate the sex specific 8-year mortality rates reported in [6] (men: 14.9%, women: 11.1%), derived from linkage to Ireland's Central Register Office records [43]. Due to access restrictions, the official mortality data were unavailable. Therefore, mortality

Table 4.2: Directional validation of the seven features accounting for 95% of gait speed reserve variance as identified by [4].

Predictor 1	**Predictor 2**	**Expected Direction / Justification**	**Observed r**
Grip Strength (GRIP)	Age	**Negative (−)**: GRIP is a positive predictor of GSR, while age is a negative predictor.	-0.265
GRIP	Chair Stand Time	**Negative (−)**: Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests.	-0.170
Montreal Cognitive Assessment (MOCA) Score	Age	**Negative (−)**: Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve.	-0.381
Education	MOCA Score	**Positive (+)**: Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR.	0.367
Body Mass Index (BMI)	Age	**Weak / Mixed**: BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak.	-0.073

status was first assigned by randomly selecting deceased participants from the wave 1 cohort aged 50 years or older to match the published sex specific mortality proportions observed at wave 5 follow up. An equal number of surviving participants were then randomly sampled within each sex to construct balanced datasets for model training. The resulting random proxy datasets included 1,114 men (557 deceased) and 982 women (491 deceased), reflecting the baseline wave 1 population aged ≥ 50 years ($n = 8,172$: 3,743 men and 4,429 women).

Two variants of this proxy were implemented:

(i) the purely random assignment described above, used as a structural baseline, and

(ii) biologically informed probabilistic assignment based on latent FR structure extracted via Principal Component Analysis (PCA). This dual construction allowed separation of statistical artefact from biologically meaningful signal.

The random proxy serves as an initial diagnostic benchmark. Mortality labels were assigned by randomly selecting deceased participants within each sex to match the published 8-year mortality proportions. While this preserves overall class balance, it does not encode any biological or FR related information and therefore serves only as a structural validation baseline.

For the biologically informed proxy, the 16 available FI items were standardised to zero mean and unit variance using z-score transformation. PCA was then performed separately within each sex on complete case observations to extract the principal component representing the dominant latent FR structure. The First Principal Component (PC1) was retained as a latent FR score, and its direction was aligned such that higher PC1 scores correspond to higher FR by enforcing a positive correlation with the calculated FI. This latent FR score was subsequently combined with age to construct a probabilistic mortality assignment reflecting both biological vulnerability and time dependent risk.

Two model variants were tested for each sex:

1. **FI only model:** using the 16 available FI items, without age.

2. **FI + Age model:** including age as an additional predictor.

Model training and evaluation followed the procedure reported in [6] as closely as permitted by the available data. Features were min-max normalised prior to modelling. For each sex, data were split into training (80%) and testing (20%) sets. To mitigate overfitting and obtain robust estimates of model performance, 5-fold stratified cross-validation was applied on the training data. AUC was selected as the primary evaluation metric due to its widespread use in prior TILDA and ML based studies (Figure 2.2) and its robustness to class imbalance.

For six of the sixteen FI items, the cleaned dataset transformations differed from the scoring rules published in [6]. To align precisely with the original 16/32-item FI methodology, these items were recoded following the published ordinal and dichotomous mappings. This step was necessary to ensure that the six remapped FI items reflects the same interpretation of health deficits used in the literature and is consistent with standard FI construction principles [17]. Models were then re-evaluated using this partial remapping FI to assess sensitivity to feature encoding.

The resulting AUC values are reported separately for (i) originally cleaned (Figure 4.3) vs 6/16 recoded FI items (Figure 4.4), (ii) random vs PCA informed mortality assignment, (iii) FI only vs FI + Age models, and (iv) men and women.

These results highlight several key points:

1. **Random proxy baseline (cleaned vs recoded FI):** Under purely random mortality assignment, discrimination remained at chance level for both versions of the FI. For the cleaned 16 FI items, the FI only model achieved AUC = 0.507 (men) and 0.512 (women), increasing marginally to 0.516 (men) and 0.531 (women) when age was included. When six of the sixteen available deficits were recoded to align with the published FI scoring rules [6], the AUC values increased to 0.567 (men) and 0.537 (women) for the FI only model, and to 0.572 (men) and 0.605 (women) for the FI + Age model. Despite this increase relative to the fully cleaned encoding, discrimination remained close to chance under random label assignment.

   These results confirm that, without embedding biological FR structure into the outcome, predictive performance reflects statistical noise rather than meaningful signal, regardless of encoding refinement.

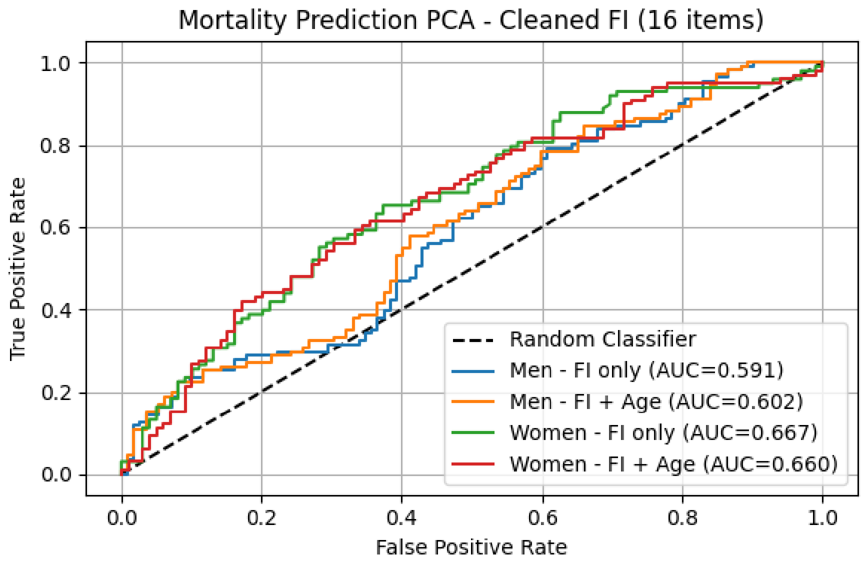

Figure 4.3: Receiver Operating Characteristic (ROC) curves for mortality prediction using 16 cleaned frailty index items under the PCA informed mortality proxy. Separate models are shown for men and women, with and without age.

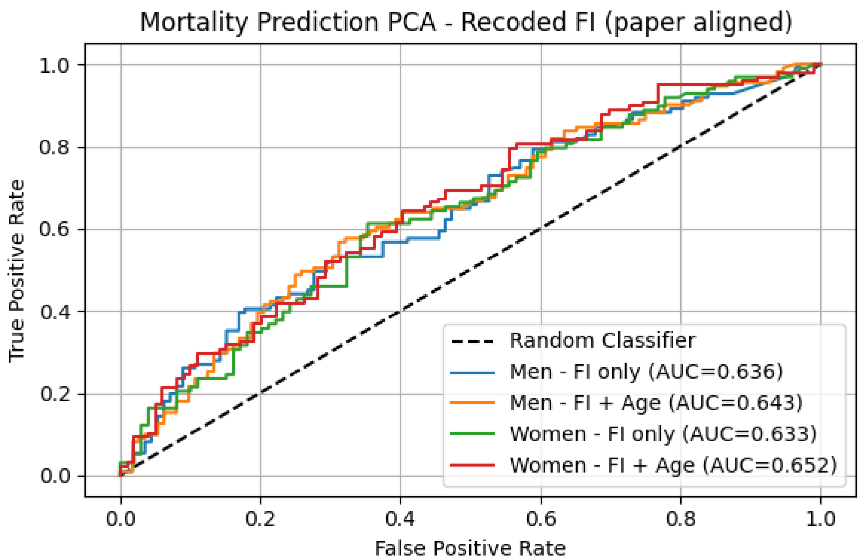

Figure 4.4: ROC curves for mortality prediction using the 6/16 recoded frailty index items aligned with [6]. Performance is shown separately for men and women, with and without age.

2. **PCA informed mortality proxy (cleaned vs recoded FI):** When mortality assignment incorporated the PCA derived FR representation PC1, discrimination improved relative to the random baseline for both FI constructions. For the 16 cleaned FI items, the FI only model achieved AUC = 0.591 (men) and 0.667 (women), increasing to 0.602 (men) and 0.660 (women) when age was included. When six of the sixteen deficits were recoded in accordance with the published scoring framework, the FI-only model achieved AUC = 0.636 (men) and 0.633 (women), while the FI + age model achieved 0.643 (men) and 0.652 (women).

   This indicates that closer alignment with published ordinal deficit scoring improves discrimination relative to the simplified numeric encoding applied during initial cleaning.

3. **Comparison with published performance:** Absolute AUC values remain below those reported in [6] (approximately 0.70 for FI only and up to 0.80–0.90 when age is included), which is expected given the use of a constructed mortality proxy and only 16 of the original 32 FI items. However, the methodological sequence reported in the original study is preserved:

   (i) Random assignment yields chance level performance,

   (ii) Extracting latent FR structure via PCA improves discrimination,

   (iii) Inclusion of age generally improved mortality prediction.

   The cleaned dataset maintains realistic sex distributions (45.8% men, 54.2% women) and mortality proportions consistent with published estimates, confirming that preprocessing preserved demographic and structural integrity required for downstream modelling.

Chapter 5

Empirical Evaluation

5.1 Modelling and evaluation framework

Building on the cleaned and validated datasets from Chapters 3 and 4, this chapter presents the empirical evaluation of ML models for predicting chronic disease onset in the TILDA cohort.

The evaluation is based on participants aged between $50-84$ years with a mean age of 66. To ensure that the prediction task reflects *incident* (new onset) disease, individuals with the target condition already present at the start of a transition were excluded, leaving only at risk observations for training and testing. This rule was applied consistently across all designs and for both Track A (Any disease outcome) and Track B (disease specific outcomes).

Final analysis samples differed by design and tracks due to data structure. The baseline design (D1) uses a single transition (Wave 1$\rightarrow$Wave 5), with a follow-up period of approximately 8 years. The pooled multi-wave designs (D2 and D3) aggregate all eligible consecutive transitions (W1$\rightarrow$W2, W2$\rightarrow$W3, W3$\rightarrow$W4, W4$\rightarrow$W5), each spanning approximately 2 years, thereby capturing shorter term dynamics.

Sample sizes and incident rates for Track A are summarised in Table 5.1. Track B, cohort sizes and incident rates vary by condition. Appendix H provides these details for each disease specific model.

5.1.1 Outcome definitions

Two complementary prediction tasks were defined.

Track A: Predicting any chronic disease onset. Track A uses a binary outcome indicating the first occurrence of any chronic disease during follow-up.

- In D1, disease free status at wave one was established using a composite of self-reported conditions (e.g., eye conditions, cardiovascular conditions, diabetes related conditions, general chronic conditions, limiting long term illness, and self-reported chronic/health problems). Incident cases were then defined using any positive ICD-10 disease codes from the WHO (*International Statistical Classification of Diseases and Related Health Problems 10th Revision*) at wave five (e.g., mental and behavioural disorders & musculoskeletal diseases).
  - No composite ICD-10 indicator exists in wave 1 to establish a clinical baseline. These clinical codes were recorded unevenly across the middle waves, with only three indicators available in wave 2, two in wave 3, and one in wave 4, before a comprehensive set of 16 variables appeared at wave 5, partially a consequence of the manual low heterogeneity filtering flow described in Section 3.4.
  - Physical & Cognitive Health (PH) variables reflect prior doctor diagnoses ("has a doctor ever told you"), whereas the "self-reported long term chronic/health problem" variable captures general perceived health burden without requiring clinical confirmation.
  - All feature scoring schemes and labels are provided in Appendix I.
- In D2 and D3, the "self-reported long term chronic/health problem" variable (where $y = 0$ denotes no chronic illness and $y = 1$ denotes presence of chronic illness) was used to both define the outcome and identify the at risk cohort (those with $y = 0$ at transition start) for inclusion in each wave. This variable is available across all waves, enabling consistent application

across all four consecutive transitions.

$$\text{Coverage}_y = \frac{\sum_{i=1}^{N} \mathbb{1}(y_i \neq missing(-1))}{N} \times 100 = 99.91\%$$

- An analysis of wave 5 data compared long term chronic/health problem against clinical records. While raw agreement reached 59.5%, (*Cohen's Kappa for agreement beyond chance*) ($\kappa = 0.167$) indicates poor agreement, showing that these variables capture different aspects of health status. Among participants with a confirmed clinical diagnosis, 45% reported themselves as healthy, whereas 38% of those without a clinical diagnosis reported a long term chronic/health problem. This low agreement suggests that "self-reported long term chronic/health problem" and clinically coded disease are distinct but complementary constructs, one capturing perceived illness burden and the other capturing objectively documented diagnoses.

Table 5.1: Sample sizes and incidence rates by design for Track A.

Design	**Observations (n)**	**Incident cases (n)**	**Incidence rate (%)**
D1: Baseline (W1→W5)	916	248	27.07
D2: Multi-wave pooled	14,296	3,212	22.5
D3: Top-20 (reduced)	14,296	3,212	22.5

Track B: Disease specific prediction. Track B targets individual chronic disease outcomes, e.g., circulatory disease, diabetes, and functional decline. For each prediction task, the outcome was defined as the first recorded diagnosis of the specific condition, and participants were excluded if the condition was present at the baseline of the corresponding transition. Scoring details for each disease specific outcome are also provided in the same Appendix I. Additionally, a predictor add-on analysis was conducted to evaluate whether individual candidate features from domains beyond the top-20 set contribute independent predictive value.

5.1.2 Modelling designs

Four supervised learning algorithms were evaluated across all main Track A designs (D1, D2, D3) to compare linear and nonlinear model families under the same evaluation framework.

1. **LR** — An interpretable linear baseline consistent with its established use in TILDA mortality and disease prediction [35], providing coefficients and odds ratios directly. In D1 and D2, a fixed configuration was used: `solver="liblinear"`, `C=0.001`, `class_weight="balanced"`, `max_iter=2000`. In D3, hyperparameters were tuned via `RandomizedSearchCV` over `C` (log-uniform $[10^{-4}, 10]$), `solver=["saga"]`, `l1_ratio` (uniform $[0, 1]$), and `max_iter` $\in \{1000, 2000, 5000\}$.

2. **XGB** — A gradient boosted tree ensemble selected for its demonstrated performance in TILDA diabetes prediction using multidomain predictors [19], capturing nonlinear interactions in heterogeneous tabular health data. In D1 and D2, fixed parameters were used: `n_estimators=1500`, `max_depth=3`, `learning_rate=0.03`, `min_child_weight=5`, `gamma=0.1`, `subsample=0.8`, `colsample_bytree=0.7`, with class imbalance handled via `scale_pos_weight` (negative to positive ratio). Early stopping (`early_stopping_rounds=50`) on an internal 20% validation split determined the optimal number of boosting rounds. The model was then refit on the full training set at that iteration. In D3, all tree and regularisation parameters were additionally tuned via `RandomizedSearchCV`.

3. **RF** — A bagging based ensemble with established use in falls, frailty, and functional decline prediction within TILDA [6, 20], providing robustness through random feature subsampling (`max_features="sqrt"`). Fixed parameters in D1/D2: `n_estimators=500`, `max_depth=5`, `min_samples_leaf=10`, with class imbalance handled via `class_weight={0:1, 1:neg/pos ratio}`. In D3, `n_estimators` $(100 - 800)$, `max_depth` $(3 - 10)$, `min_samples_leaf` $(5 - 30)$, and `max_features` $\in$ {`"sqrt"`, `"log2"`} were tuned via random search.

4. **MLP** — A feedforward neural network with two hidden layers, included to represent the neural family in the four model comparison. Neural networks have been proposed as a promising tool for clinical risk prediction due to their ability to capture hierarchical nonlinear patterns [44]. Their inclusion here tests directly whether this advantage holds in survey based longitudinal ageing data against the tree-based baselines. Fixed parameters in D1/D2: `hidden_layer_sizes=(32,16)`, `activation=''relu''`, `solver=''adam''`, `alpha=0.01`, `learning_rate_init=0.001`, `max_iter=500`, with early stopping (`n_iter_no_change=20`, `validation_fraction=0.1`). Class imbalance was handled via `compute_sample_weight(class_weight=''balanced'')` passed at fit time, as MLP does not natively support `class_weight`. In D3, `hidden_layer_sizes` $\in \{(32,), (64,), (32{,}16), (64{,}32), (128{,}64)\}$, `activation` $\in$ {`''relu''`, `''tanh''`}, `alpha` (log-uniform $[10^{-4}, 1]$), and `learning_rate_init` (log-uniform $[10^{-4}, 10^{-1}]$) were tuned via random search.

The preprocessing backbone median imputation (`SimpleImputer`), variance filtering (threshold = 0.01), and missingness filter (> 50% on training only) was shared across all models, with scaling (`StandardScaler`) applied only to LR and MLP (sensitive to feature magnitude). Tree-based models were trained on the raw imputed features without scaling. All steps were fit exclusively on training data (prior to internal train/validation splitting) before being applied to validation and test sets, preventing information leakage.

Track B models used XGB exclusively, as it achieved the strongest and most consistent performance across D2 and D3 in Track A (see Section 5.2). The same model configuration and hyperparameter search space used in D3 was applied in Track B.

The three primary modelling designs were implemented to address and assess the effect of baseline prediction, longitudinal pooling, and feature reduction with optimisation, respectively.

- **D1 (Baseline):** A single W1→W5 transition. Only participants disease free at wave 1 (using self-reported and derived disease variables, plus explicit ICD-10 source variables) were included, with incident disease defined by any positive ICD-10 coded diagnosis at wave 5 ($n = 916$, incidence 27.1%). Feature selection used `SelectKBest` ($k = 20$) and CV used `StratifiedKFold` (one observation per participant).

- **D2 (Multi-wave pooled):** All four eligible consecutive transitions (W1→W2, W2→W3, W3→W4, W4→W5) were stacked into a single pooled dataset. Disease free status and incident outcome were both defined using the self-reported long term chronic/health problem variable at each transition. A numeric wave transition indicator was added as a feature to capture temporal context. The pooled cohort comprised $n = 14{,}296$ person wave observations (incidence 22.5%). Feature selection used `SelectKBest` ($k = 100$), and CV used `StratifiedGroupKFold` to ensure no participant appeared in more than one fold.

- **D3 (Top-20 + Hyperparameter Optimization (HPO)):** Built directly on the D2 cohort and wave structure. The feature set was restricted to the 20 highest ranked predictors by XGB gain importance from D2, plus the wave transition indicator. No additional feature selection step was applied. HPO was performed for all four model families using `RandomizedSearchCV` (200 iterations) within a `StratifiedGroupKFold` framework.

All three designs shared the same participant level train/test split (80%/20%, stratified by outcome), the same missingness filter and variance threshold applied on training data only, and the same five fold CV evaluation protocol. Track B models followed either the D2 or D3 structure depending on the outcome being predicted. Full details are provided in Subsection 5.2.4.

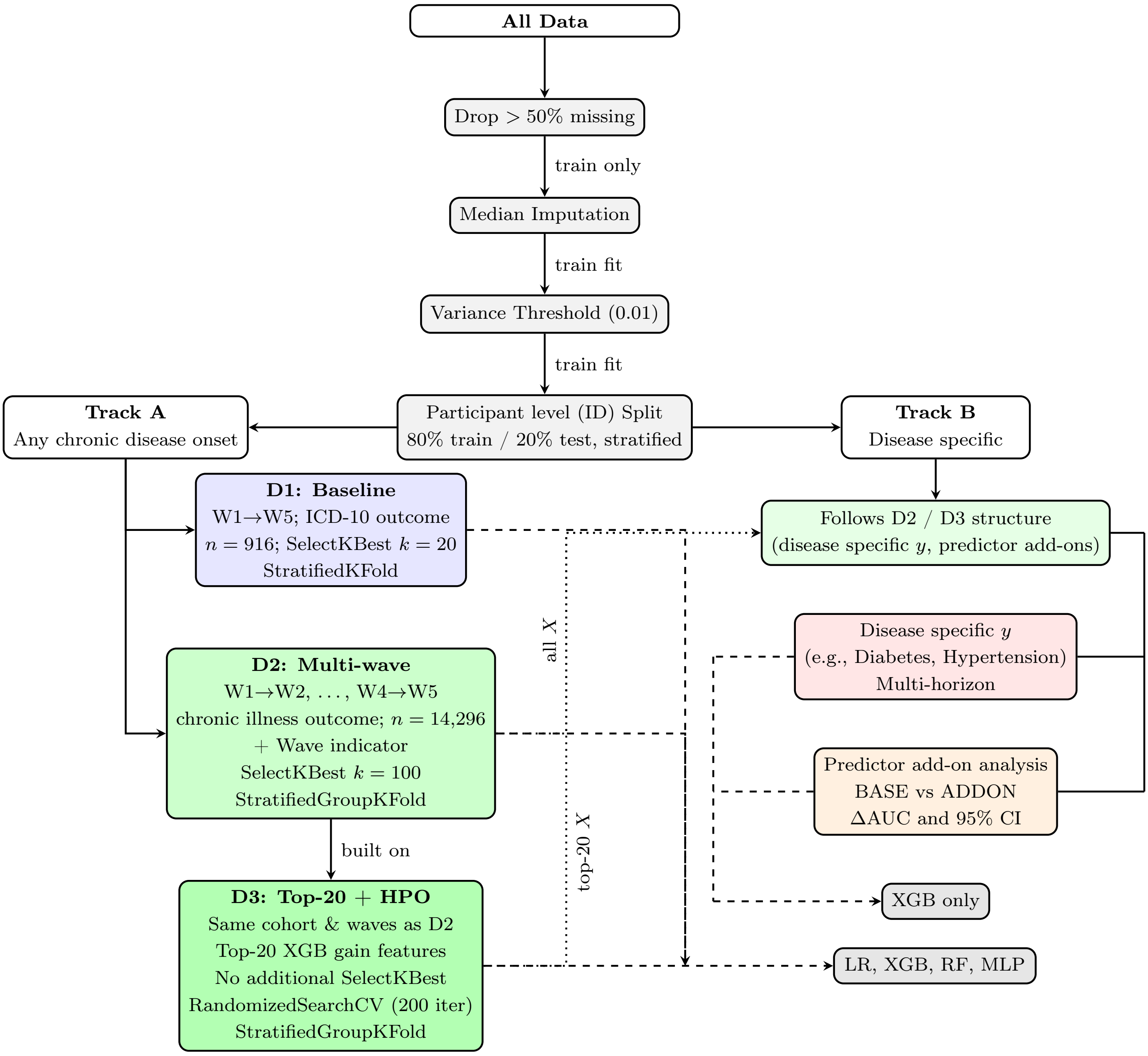

Figure 5.1: Pipeline overview: shared preprocessing applied to all data before branching into Track A (any chronic disease, D1/D2/D3) and Track B (disease specific, following D2/D3 structure). D3 is built directly on D2's cohort and wave structure, adding feature reduction and hyperparameter optimisation.

Figure 5.1 notes:

(i) All grey pipeline steps (missingness filter, imputation, variance threshold, split) are shared across Track A and Track B. All fitting occurs on training data only.

(ii) D1 uses a single W1→W5 transition with `StratifiedKFold`, D2 and D3 use all four pooled transitions with `StratifiedGroupKFold` to prevent participant level leakage across folds.

(iii) D3 is built on D2: it uses the same cohort, waves, and wave indicator, but replaces `SelectKBest` with the top-20 features ranked by XGB gain from D2, and adds HPO for all model families.

(iv) Feature counts after the missingness filter: D1 retains ≈570 features, further reduced to $k = 20$ by `SelectKBest` before model training. D2 retains ≈261 features, reduced to $k = 100$. D3 uses exactly 21 features (top-20 + wave indicator) with no further selection.

(v) Track B uses XGB only, following the D2 or D3 design with a disease specific outcome and explicit exclusion of direct outcome proxies from the predictor set.

5.1.3 Participant level data splitting and leakage prevention

To prevent identity based leakage where models memorise participant specific health signatures rather than learning general risk patterns, all splitting was performed at the **participant level**. Individuals were assigned exclusively to either the training, validation, or test set, ensuring no participant contributed data to more than one split. The overall dataset was first split into 80% for model development and 20% as a held-out test set. The 80% development partition was then further divided 80%/20% internally, yielding a final three way allocation of 64% training, 16% validation, and 20% test. Splitting was performed using participant identifiers to ensure group wise separation, with stratification by outcome to preserve class balance across splits. As demonstrated in longitudinal brain Magnetic Resonance Imaging (MRI) analysis [45], participant level splitting prevents identity confounding, where models learn individual health signatures rather than generalisable risk patterns.

D1 used `StratifiedKFold` ($k = 5$). D2 and D3 used `StratifiedGroupKFold` ($k = 5$), ensuring all observations from a given participant are assigned to a single fold and no participant appears in both training and validation sets simultaneously. Three additional leakage controls were applied across all designs:

- **Pipeline isolation:** Imputation medians, variance masks, and feature selection scores were computed on the training partition only. The resulting transformations were then applied unchanged to validation and test data.

- **Temporal ordering:** Predictors were drawn exclusively from the baseline wave of each transition (wave n) to predict outcomes at the subsequent (wave $n + 1$), eliminating any look ahead bias.

- **Exclusion of direct proxies:** Variables that directly encode the outcome being predicted (e.g., self-reported disease source indicators that feed into the ICD-10 codes used as outcome, such as ("has a doctor ever told you that you have") a Ministroke or TIA and Parkinson's disease for nervous system diseases), were removed from the predictor set prior to modelling.

5.1.4 Evaluation protocol and metrics

Model performance was assessed using discrimination, calibration, and classification metrics. The primary metric was AUC. Models requiring HPO (D3) used `RandomizedSearchCV` within the same CV framework.

Calibration was evaluated using the Brier score, which measures the accuracy of predicted probabilities, while classification performance was summarised using the F1 score, capturing the balance between precision and recall in the presence of class imbalance (e.g., $22 - 27\%$ incidence). Additional metrics, including sensitivity, specificity, and confusion matrices, were used to support interpretation.

For Track B predictor add-on analysis, 95% Confidence Interval (CI)s for performance deltas were estimated via bootstrap resampling to ensure statistical significance.

5.2 Main results

Performance is evaluated first across the three Track A designs (D1, D2, and D3), followed by cross design and model comparisons. Results conclude with Track B disease specific outcomes.

5.2.1 Track A: Any disease onset

Results are presented across the three modelling designs, with tables summarising discrimination (AUC), calibration (Brier score), and classification performance (F1), alongside training, validation, and test metrics to assess generalisation.

5.2.1.1 Design 1 performance

Table 5.2 summarises the performance of the four models on the single transition (Wave 1 $\rightarrow$ Wave 5). Discrimination is weak with test AUC values ranging from 0.53–0.58, and the larger train-test gaps for tree based models indicate stronger overfitting in the baseline design.

Table 5.2: Predictive performance across model families for D1 (baseline design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.589	0.677	0.582	0.571	0.245	0.411
XGBoost	0.569	0.825	0.563	0.544	0.240	0.398
Random Forest	0.579	0.871	0.565	0.534	0.231	0.440
MLP	0.536	0.664	0.552	0.582	0.241	0.419

5.2.1.2 Design 2 performance

Table 5.3 shows the results when using all consecutive transitions (pooled multi-wave design). All models improve substantially compared to D1 5.2.1.1, with test AUC values ranging from 0.69 to 0.72.

Table 5.3: Predictive performance across model families for D2 (multi-wave design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.715	0.728	0.715	0.709	0.217	0.460
XGBoost	0.716	0.736	0.719	0.715	0.214	0.468
Random Forest	0.711	0.740	0.717	0.711	0.212	0.452
MLP	0.698	0.749	0.706	0.699	0.208	0.455

5.2.1.3 Design 3 performance

Table 5.4 presents performance using only the 20 most informative predictors (derived from XGB model trained on D2 5.2.1.2) with HPO applied across all models. Discrimination remains effectively unchanged compared to D2, with test AUC values tightly clustered around 0.71.

Table 5.4: Predictive performance across model families for D3 (tuned design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.716	0.718	0.717	0.712	0.212	0.459
XGBoost	0.718	0.736	0.719	0.713	0.209	0.462
Random Forest	0.717	0.766	0.718	0.713	0.205	0.462
MLP	0.713	0.743	0.710	0.712	0.207	0.467

5.2.2 Cross design comparison

The best test AUC increased from 0.582 in D1 5.2.1.1 (achieved by MLP) to 0.715 in D2 5.2.1.2 by XGB, representing an improvement of 0.133, demonstrating the benefit of incorporating pooled longitudinal transitions.

Reducing the feature set to the top-20 predictors in D3 maintained this gain, with XGB achieving a test AUC of 0.713, only 0.002 lower than D2, confirming that most of the predictive signal is concentrated in a small subset of variables, and that performance is robust to feature reduction.

Across designs, CV performance also improves and stabilises:

- D1: 0.589 ± 0.072 (LR)

- D2: 0.716 ± 0.007 (XGB)

- D3: 0.718 ± 0.006 (XGB)

The reduction in standard deviation from 0.072 (D1) to 0.007 (D2) is not only a statistical artifact of a larger sample. It reflects a shift in the nature of the learning problem. D1 asks the model to learn from a single snapshot of health at wave 1 and predict a clinically coded outcome eight years later, a task where signal is sparse and the outcome is heterogeneous, and chance variation dominates. D2 pools 14,296 person wave observations, each capturing a tighter 2-year transition window in which self-reported health burden is the direct and proximal outcome. This alignment between predictor and outcome horizon is the structural reason for the performance gain, not simply the larger n. The implication is that for short term incident prediction in survey cohorts, self-reported health state is a stronger signal source than objective biomarkers measured years earlier.

5.2.3 Model comparison

Once temporal structure is introduced (D2 and D3), model differences narrow: test AUC ranges shrink from 0.048 (D1) to 0.022 (D2) to 0.006 (D3), with all models clustering in the range 0.69–0.72 and no single model class dominating. XGB achieved the highest test AUC in both D2 (0.715) and D3 (0.713), though the margin over LR was narrow (≤ 0.006). The fact that the linear LR remained consistently competitive (0.709 in D2 5.2.1.2 and 0.712 in D3 5.2.1.3) suggests that risk signals in the ageing population are primarily linear. MLP slightly underperforms in D2 5.2.1.2 (0.699) but matched other models in D3 5.2.1.3 (0.712), indicating improved performance when the feature space is reduced to the most informative predictors. This indicates that individual burden indicators, particularly medication count and pain, dominate the risk score with largely linear effects, leaving limited room for nonlinear algorithms to outperform a well regularised linear baseline. The convergence of model families in D3, where the AUC range narrows to just 0.006, is a further sign that the available signal has been largely captured by the top-20 features. Improving performance from here would require richer data rather than more complex algorithms.

Figure 5.2 compares generalisation gaps (Train−Test) AUC across all three designs. The average train-test AUC gap (across all four models) drops from 0.202 in D1 to 0.030 in D2 and 0.028 in D3 – a reduction of over 85%. This demonstrates that the pooled multi-wave design effectively reduces overfitting. D1 exhibits the largest gaps, consistent with weaker signal and higher variance in estimates. In contrast, D2 and D3 show much smaller gaps, indicating improved stability. Tree based models (XGB and RF) exhibit the largest gaps in D1, consistent with the small sample size. In D2 and D3, XGB achieves the smallest gaps (0.021 and 0.023 respectively), while RF shows comparatively larger gaps (0.029 and 0.053). The strong performance of XGB in D3 is expected, as the top-20 feature set was selected using XGB gain importance from D2. Nevertheless, XGB also generalises well in D2, suggesting that its boosting mechanism is effective at learning stable patterns from pooled longitudinal data. In contrast, RF may be more prone to overfitting even with the pooled design, possibly due to its reliance on bagging and random feature selection, which may capture informative predictors less efficiently than boosting.

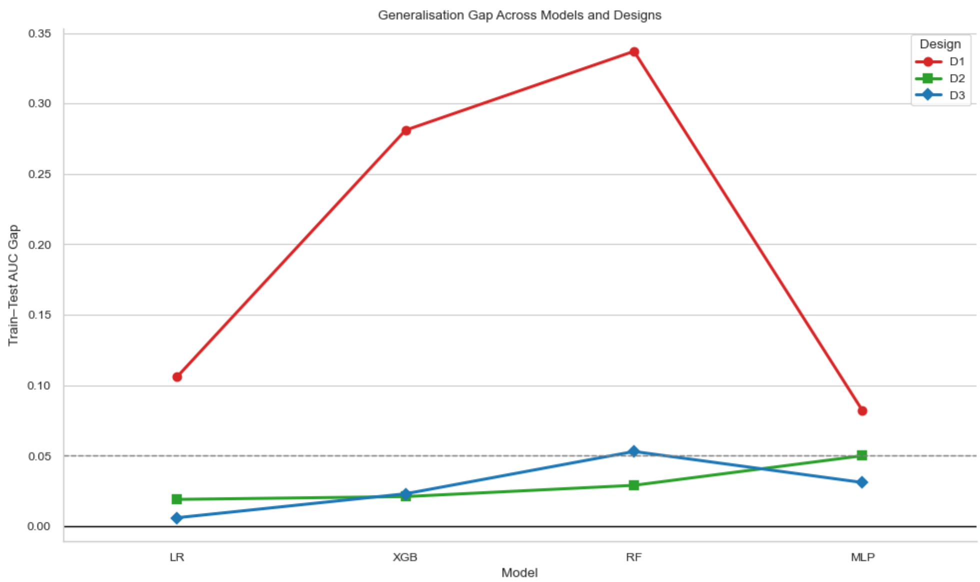

Figure 5.2: Train−Test generalisation gaps across models and designs. 0.05 marks a conventional heuristic threshold for acceptable error in AUC. Gaps below this level indicate that the model generalises well to unseen data.

The figure also highlights that model level differences in generalisation gap are more pronounced in D1 5.2.1.1, whereas D2 5.2.1.2 and D3 5.2.1.3 exhibit convergence across model families, suggesting that data structure plays a larger role than algorithmic choice in controlling overfitting.

5.2.4 Track B: Disease specific

Track B targets individual chronic disease outcomes using the same pooled cohort structure and participant level splitting as D2 and D3, with XGB as the sole classifier given its consistently strong performance in Track A. Twenty-four disease specific outcomes were evaluated using two feature strategies: all eligible predictors retained after the 50% missingness filter (for example, 261 features for functional decline and 351 for circulatory disease), and a restricted top-20 set derived from XGB gain importance.

Outcomes span key domains of ageing related health in TILDA: cardiovascular conditions, functional decline, mental health, and metabolic disease.

5.2.4.1 Outcome categories

Four outcome categories were evaluated:

- **Clinical (ICD-10):** Circulatory, endocrine/metabolic, eye, and musculoskeletal diseases (comprehensive from wave 5 only, source predictors excluded to prevent leakage).

- **Functional:** Functional decline, falls (Inc. balance), pain, hearing loss, and urinary incontinence (self-reported, available across multiple waves e.g., W2→W4, W3→W4→W5 and W2→W4).

- **Family history:** Reports of family member diagnoses across eight conditions. These are not incident diagnoses of the participant and represent a distinct prediction target: estimating the likelihood of a participant newly reporting a family history item at follow-up. These variables were retained through the manual filtering process described in Section 3.4.

- **Medication/procedure proxies:** Antidepressant, antihypertensive, cholesterol medications, doctor confirmed hypertension, and cardiac procedure. These are downstream markers of clinical decision making used as prediction targets. They capture risk signals linked to eventual treatment rather than raw disease onset. As noted in Appendix H, incidence rates for medication proxies vary widely, 1.4% for cardiac procedure to 41.6% for doctor confirmed hypertension diagnosis (though this is based on a very small cohort of $n = 250$).

5.2.4.2 Overall performance

Figure 5.3 plots train AUC against test AUC for all disease specific models, coloured by outcome category. The diagonal represents perfect generalisation (train equals test), while dashed lines at 0.5 on both axes mark chance level performance. Points close to the diagonal indicate stable learned signal, whereas points far above it, with much higher train than test AUC, indicate overfitting. Overall, test AUC values range from approximately 0.45 to 0.72, reflecting substantial variation driven mainly by outcome type and incidence rate rather than model choice.

Functional and symptom outcomes were the most reliable across both feature strategies. Functional decline ($n = 14{,}296$, incidence 22.5%) achieved a test AUC of 0.713 with a small gap between train and test. This outcome uses the same self-reported chronic health problem variable as D2 and D3 in Track A, and the consistent performance across both tracks confirms the stability of this predictor outcome relationship and persistent FR signal identified in the literature. FOF ($n = 18{,}114$, incidence 14.0%) achieved 0.721, the highest test AUC, closely followed by antihypertensive prescription, reflecting the well documented link between functional limitation and falls risk in TILDA. Hearing loss ($n = 10{,}780$, incidence 15.1%) reached 0.678, and urinary incontinence ($n = 16{,}208$, incidence 8.4%) achieved 0.682, both with gaps below 0.10. These outcomes benefit from consistent measurement across waves, sufficient event counts, and clear physiological pathways, which support more stable longitudinal prediction.

Medication proxies achieved competitive test AUC values. Antihypertensive prescription (n = 19,845, incidence 4.5%) reached 0.718, and antidepressant prescription (n = 4,378, incidence 3.2%) reached 0.702, despite low incidence. Both models showed train AUC above 0.99, indicating overfitting to training specific patterns rather than fully generalisable risk patterns. The test results show that the models can discriminate between cases and non cases, although part of this performance likely comes from established treatment profiles. For example, a participant already receiving antihypertensives at baseline has a clinical profile strongly associated with future prescription continuation. These models are therefore informative for risk stratification but should not be interpreted as predicting new disease onset.

Clinical coded ICD-10 outcomes showed moderate performance. Circulatory disease (n = 14,539, incidence 12.2%) and eye diseases (n = 4,396, incidence 10.1%) were the strongest, reaching test AUC values of 0.626 and 0.653, respectively. This aligns with findings that cardiovascular outcomes in TILDA are supported by well established clinical predictors. Musculoskeletal diseases (n = 4,269, incidence 6.5%) and endocrine and metabolic diseases (n = 4,117, incidence 8.8%) performed near chance, with test AUC values of 0.507 and 0.515 under the all features strategy, alongside train AUC values of 1.000 and 0.967. This pattern indicates that the models memorised training labels without learning signal that transfers to new data. This issue is discussed further in Section 5.2.4.4.

Family history models produced the most variable and least reliable results. Using all features, test AUC ranged from 0.513 for cancer (n = 3,971, incidence 13.4%) to 0.665 for high cholesterol (n = 4,312, incidence 12.0%), while training AUC reached 1.000 for several outcomes such as Alzheimer's disease and cancer, indicating severe memorisation. Switching to the top-20 feature set consistently degraded performance, with several models dropping below 0.50. The root cause is the indirect nature of the target: family history variables capture a participant's reported knowledge of a relative's diagnosis, rather than an outcome experienced by the participant. As a result, there is no direct causal pathway linking participant level predictors to these outcomes, limiting the model's ability to learn generalisable patterns. What is learned from the training data therefore does not transfer well to unseen data.

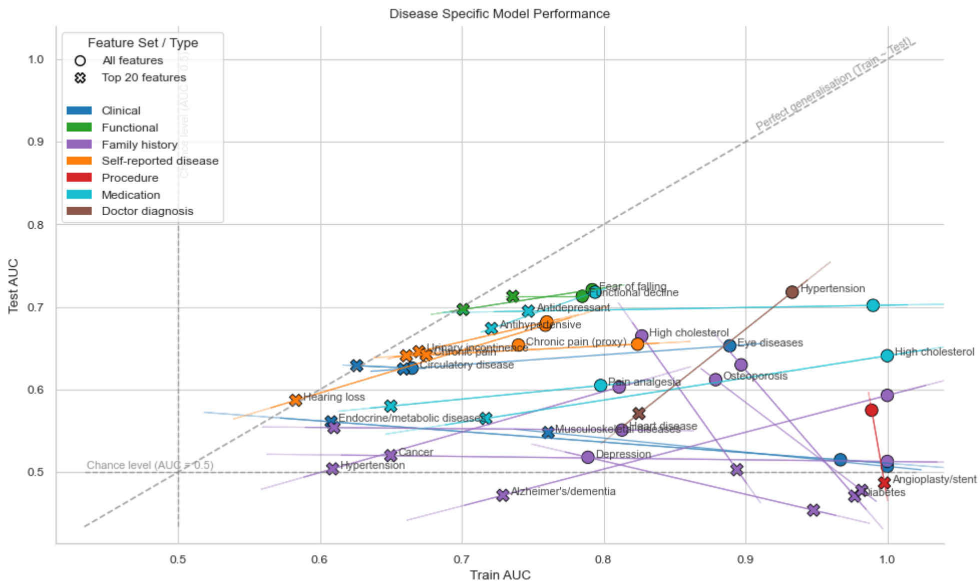

Figure 5.3: Train AUC vs Test AUC for all Track B disease specific models, coloured by outcome category. Circle markers represent all feature models and cross markers represent top-20 feature models, with a line connecting the two feature strategies per disease. The diagonal dashed line marks perfect generalisation (Train ≈ Test). Dashed lines at 0.5 on both axes indicate chance level discrimination. Confidence ellipses illustrate within category spread.

5.2.4.3 Incidence and predictability

Cohort sizes and incidence rates for each outcome are provided in Appendix H. Incidence rate appears to be the most consistent driver of model stability in Track B. Outcomes with incidence above approximately 12%, particularly when measured consistently across waves, tended to achieve test AUC values above 0.65. In contrast, outcomes with incidence below approximately 7% generally produced near chance test AUC and large gaps between train and test regardless of feature strategy.

A model trained on few positive examples cannot learn stable decision boundaries. Class imbalance weighting via `scale_pos_weight` partially compensates, but when positive cases are very sparse, the model can achieve high training scores by memorising those few cases without generalising. This is why Train = 1.000 appears for musculoskeletal diseases, Alzheimer's disease, and cancer in the all features models, none of which exceed test AUC of 0.54 despite high training performance.

Doctor confirmed hypertension is an outlier worth noting. Its incidence rate of 41.6% is the highest in Track B, and it achieved test AUC of 0.718. However, the cohort is very small at $n = 250$, making this estimate less reliable. This result should be interpreted with caution, as a cohort of this size can produce highly variable test scores depending on the specific participants allocated to the test split.

5.2.4.4 Overfitting and feature reduction

For outcomes where the all features strategy produced train AUC substantially above 0.90 alongside near chance test AUC, the top-20 strategy provides a more reliable estimate of generalisable performance, even when absolute performance remains modest. Musculoskeletal diseases moved from train 1.000 / test 0.507 to train 0.761 / test 0.548 under feature reduction. Endocrine and metabolic diseases similarly improved from 0.515 to 0.561. These are not large gains, but they indicate a more calibrated model that has learned from signal rather than noise. In contrast, for outcomes where both strategies produced similar test AUC with small gaps, such as functional decline, FOF, and hearing loss, the feature reduction confirms stable learning rather than addressing overfitting.

For family history targets, feature reduction worsened rather than helped. With only 20 features, the model is further constrained in its ability to identify even proxy signal for family reported diagnoses. This is consistent with the broader point that the target type, rather than the feature count, drives instability in these models.

The overall pattern across Track B is consistent with the preprocessing decisions made in Section 3.4. Variables were excluded during the manual filtering stage if they exceeded the missingness threshold, exhibited constant values, or had a single response category dominating more than approximately 90% of observations. This process improved overall signal quality, but it also reduced the number of available predictors for outcomes that are inherently sparse or indirectly measured. Outcomes with rich, well distributed predictor signals, such as functional decline and FOF, benefited most. Outcomes dependent on rare or indirect signals, such as Alzheimer's family history or cardiac procedure, remained constrained by both the target definition and the reduced predictor space. The results in Figure 5.3 should therefore be interpreted not only as a function of modelling choices, but as a reflection of data availability and the structural properties of each outcome in TILDA.

5.3 Model interpretability and feature importance

Feature importance analysis across all designs reveals three consistent findings. First, predictive signal is concentrated in a small, highly redundant set of health burden indicators, primarily spanning the Physical/Clinical and Functional thematic domains defined in Section 3.3, with other domains adding little additional discriminative information beyond the baseline feature set. Second, the most important predictors shift fundamentally between designs: objective clinical biomarkers dominate the 8 year prediction window in D1, while medication burden and self-rated health dominate the 2 year transitions in D2 and D3. Third, no single domain specific predictor added statistically significant predictive value, as confirmed by the predictor add-on analysis in Section 5.3.4. Full feature importance bar plots for D1 and D2 across all four model families are provided in Appendix J. Figure 5.4 displays the add-on Δ visualisation.

Global SHAP beeswarm plots for D1, D2, and D3 (all four model families), together with four disease specific XGB models from Track B, are provided in Appendix K. In D1, blood pressure, anthropometric markers, and social activity attendance carry the highest mean |SHAP| values, consistent with the 8-year ICD-10 prediction task. The binary hypertension indicator ranks fourth in LR and MLP ($|\text{SHAP}| \approx 0.025$) but receives exactly zero weight under XGB, which resolves the same information through continuous blood pressure splits. Social activity attendance appears in the D1 top 5 across all models but is absent from D2 entirely, a temporal specificity consistent with the literature reviewed in Section 2.3: participation in community activities carries long-horizon predictive information that does not translate to 2-year self-reported transitions. In D2, medication count produces the most concentrated rightward shift under XGB

(|SHAP| = 0.194), while self-rated health leads for LR (0.118), reflecting how linear and tree-based models differently weight these interacting predictors. University of California Los Angeles (UCLA) loneliness appears at rank 16 in the LR beeswarm (|SHAP| = 0.034) but is absent from the tree and neural models, which explains the near-zero add-on delta in Section 5.3.4: a weak linear signal exists but is absorbed by correlated pain and healthcare utilisation predictors in ensemble models.

The Track B plots reveal outcome specific predictor structures that diverge sharply from the Track A pattern, making visible the measurement gap between self-reported and clinically coded disease documented in Section 5.3. Functional decline replicates the Track A burden hierarchy (self-rated health 0.172, medication count 0.166, chronic pain proxy 0.137). FOF is driven instead by age (0.232), physical impairment (0.207), and two distinct anxiety measures (0.110, 0.071), none of which appear in the Track A top 20, consistent with the cognitive affective dimension of fear of falling [30]. For hearing loss, the pre-existing hearing difficulty indicator alone accounts for |SHAP| = 0.367, more than twice the next feature, confirming the model predicts progression of existing deficit rather than genuinely incident cases. Circulatory disease is age dominated (0.150), with retirement status (0.031) entering the top 5. Across Track B, the signal sources shift with the biology of each condition, which is why no single compact predictor set generalises across all disease specific targets.

5.3.1 Clinical biomarkers dominate

In D1, the top-20 features selected by `SelectKBest` were dominated by cardiovascular and anthropometric measurements from the wave 1 health centre assessment (see Appendix J.1). Blood pressure variables, covering seated and standing systolic and diastolic readings and their means, appeared prominently across all four model families. Anthropometric FR markers, specifically hip circumference (0.084 for RF, 0.037 for MLP), waist circumference, and BMI, were consistently ranked alongside trail making test time as a marker of cognitive processing speed. Social engagement through attendance at classes or lectures, health insurance coverage, mother's age at death as a familial longevity signal, socioeconomic group, and physical impairment count contributed additional signal across models.

This pattern aligns with what the literature review chapter identified as the primary drivers of chronic disease and mortality in this cohort. Blood pressure is a well established cardiovascular risk marker [35], anthropometric measures connect to the FR and obesity pathway in TILDA [29], and trail making test time reflects the cognitive reserve framework discussed in Chapter 2 [21]. The dominance of these objectively measured variables reflects the availability of a full wave 1 health centre assessment providing objective clinical measurements that were less consistently available in later waves.

Across model families, the ranking order varied modestly. Seated systolic blood pressure ranked first for both XGB (gain 0.072) and RF, which is notable given that these are both tree based models using different importance metrics, suggesting that the feature is consistently identified as important across tree based approaches. By contrast, LR showed a broader distribution placing attending classes or lectures at rank one (0.039), and MLP placed hip circumference first (0.037). Importance values are not directly comparable across model families as each uses a different scoring method, so the cross model consistency in which features appear, rather than their specific scores, is the more meaningful observation here.

5.3.2 Medication burden and self-rated health emerge

Shifting from D1 to D2 produced a major change in the importance landscape (see Appendix J.2). Blood pressure variables, which dominated D1, did not appear in the top-20 for any model in D2. Medication burden and self-rated health became the primary drivers instead.

Medication count excluding supplements ranked first in XGB (0.068), RF (0.115), and MLP (0.012) in D2. Polypharmacy indicators, total medication counts including supplements, and extended medication burden measures collectively occupied much of the top 10 across all four model families. This shift reflects the change in prediction target from ICD-10 coded hospitalisation over 8 years (D1) to self-reported chronic illness over 2 year transitions (D2). Over shorter windows, medication burden is a closer indicator of health deterioration: a participant already taking multiple medications at baseline is substantially more likely to report chronic illness at the next wave.

This reflects the difference in how far ahead each design is predicting. Over 8 years, clinical measurements such as blood pressure reflect long-term biological risk that takes time to appear as diagnosed disease. Over 2 years, medication burden and self-rated health are much more direct markers of near term change. The best predictors, in short, depend on the prediction window.

Self-rated general health ranked first for LR (0.150) and appeared in the top six for all model families. This predictor is one of the most consistently reported predictors of subsequent morbidity and mortality in older adult cohorts [6], and its prominence here validates the predictive framework. Comparative self-rated health, capturing how participants assess their health relative to peers of the same age, also appeared in the top-20 across LR, XGB and RF models.

Pain related variables contributed substantially in D2. Chronic pain limiting activities, pain severity, and any chronic pain indicator all ranked consistently across model families. Chapter 2.6 identified chronic pain as a marker of underlying functional decline and a strong predictor of subsequent disease burden [16]. Functional disability items, specifically difficulty getting up from a chair after sitting, chair rising steadiness, and aggregate physical limitation counts, rounded out the top predictors. Healthcare utilisation variables, general practitioner visit frequency and outpatient visit frequency, also appeared, reflecting that participants who engage heavily with the health system consistently showed elevated risk of incident chronic illness at the next wave.

5.3.3 Cross model agreement and feature set

A notable finding across D2 is the high degree of agreement between model families regarding which features matter. The top five features by XGB gain, RF Gini importance, LR absolute coefficient magnitude, and MLP permutation importance show substantial overlap in the highest ranked predictors, with only minor ordering differences. This agreement across four different model types strengthens confidence in the results. When a linear model, two tree ensembles, and a neural network independently converge on the same top features, it is unlikely to be a peculiarity of any single method. The health burden cluster is not just one way to interpret this data, it is the dominant signal in this cohort.

The D3 and Track B top-20 feature set was derived directly from XGB gain importance in D2 and applied unchanged across all four D3 model families and all Track B disease specific models. Table 5.5 lists this feature set with thematic domain assignments following the domains described in Section 3.4.

Table 5.5: Top-20 features from XGBoost gain in D2, applied in D3 and Track B.

Feature label	Description	Thematic domain
Reported medications (excl. supps)	Total reported medications excluding supplements	Physical/Clinical
Chronic pain (proxy)	Frequency of pain experienced	Functional/Symptom
Polypharmacy (excl. supps)	Taking five or more medications excluding supplements	Physical/Clinical
Regular medications	Number of regular medications including supplements	Physical/Clinical
Reported medications	Number of reported medications including supplements	Physical/Clinical
Self-rated health	General self-rated health (excellent to poor)	Physical/Clinical
Number of physical limitations	Count of activity limitations from mobility items	Functional
Health limits activities	Health problem limits kind or amount of activities	Functional
Pain limits activities	Chronic pain restricts daily activities	Functional/Symptom
GP visits (12 months)	Number of GP visits in past 12 months	Physical/Clinical
Large muscle ADL/IADL index	Composite of large muscle daily activity items	Functional
Pain severity	Level of pain reported	Functional/Symptom
Flu vaccination	Had a flu vaccination in past 12 months	Lifestyle
Walking/standing steadiness	Steadiness when walking or standing	Functional
Chronic pain	Any chronic pain present	Functional/Symptom
Chair rising steadiness	Steadiness when getting up from a chair	Functional
Hospital visits (12 months)	Number of outpatient visits in past 12 months	Physical/Clinical
Getting up from chair difficulty	Difficulty getting up from a chair after sitting	Functional
Functional difficulty	Difficulty stooping, kneeling, or crouching	Functional
Health vs peers	Self-rated health compared with others of same age	Physical/Clinical

'Chronic pain (proxy)' is a derived variable from CAPI including proxy respondents. 'Chronic pain' is a separate binary indicator from the derived variable codebook.

The feature set spans two primary thematic domains: Physical/Clinical (medication counts, self-rated health, healthcare utilisation) and Functional (mobility difficulty, pain, physical limitation), with one Lifestyle item (flu vaccination). Nutritional and metabolic biomarkers, psychological and cognitive measures, and sociodemographic variables are absent from the top-20. Cholesterol, the nutritional biomarker that did enter the add-on analysis, showed no statistically significant independent contribution (ΔAUC: $[-0.011, 0.023]$), consistent with the broader null finding across domains.

5.3.4 Predictor add-on analysis

The add-on analysis evaluated the incremental contribution of individual candidate features to the prediction of functional decline using a *BASE* versus *BASE+ ADDON* framework. For each candidate, a model was trained on all available predictors except that candidate (BASE), then retrained with it included. The baseline cohort and feature set are determined dynamically for each candidate. Only waves in which both the outcome target and the candidate feature were present were used, meaning cohort size and feature count varied across candidates. For example, when a history of blackout or fainting served as the candidate feature, the analysis included waves 1 to 3, yielding 929 held-out test participants and approximately 570 features entering the model after the missingness filter. When education level served as the candidate, waves 3 to 5 were used, yielding 660 test participants and approximately 464 features. The delta in AUC and (Average Precision (AP)) was estimated with bootstrapped 95% CIs ($B = 2{,}000$ resamples on the held-out test set). Full delta results are shown in Figure 5.4. The candidates span lifestyle, socioeconomic, cognitive, nutritional, and social domains, including features from thematic domains listed in Table 3.2 that are not represented in the top-20 feature set.

The purpose of the analysis is to investigate whether any domain contributes unique predictive information beyond this already rich baseline. Given the large baseline feature sets, marginal gains from a single candidate are expected to be small. No candidate produced a statistically significant improvement across any metric. Formally, for each candidate the null hypothesis $H_0\colon \Delta\text{AUC} = 0$ was tested via bootstrapped 95% confidence intervals ($B = 2{,}000$ resamples), rejection requires the interval to exclude zero. Every interval crossed zero, providing a consistent null result across all domains. Selected results, organised by domain cluster, are as follows.

(i) **Functional predictors** (chair rising difficulty, physical limitations, chair rising steadiness, FOF). Among the functional candidates, the highest BASE AUC occurred when getting up from a chair difficulty served as the candidate feature (0.707). Adding it to the BASE produced a near zero marginal gain (CI: $[-0.012, 0.002]$). Physical limitation count (CI: $[-0.008, 0.007]$)

and chair rising steadiness (CI: $[-0.006, 0.010]$) showed similarly null deltas. As reviewed in Table 2.3, UGS, chair stand performance, and physical limitation are consistently identified as primary FR indicators in TILDA [7]. The small incremental contributions are consistent with strong correlation across functional predictors: pain, steadiness, and mobility difficulty items share their predictive variance, and none appears to contribute substantial independent information beyond the others.

(ii) **Socioeconomic predictors** (education level, household income). Education level produced the largest observed positive delta, with Test AUC moving from 0.711 to 0.719 (CI: $[-0.002, 0.018]$). The direction is consistent with the examined literature on social disadvantage [15], although the interval crosses zero and the effect cannot be confirmed statistically. Total household income produced a smaller positive trend ($[-0.004, 0.010]$). The smaller effect likely reflects alignment with healthcare utilisation and employment status predictors already in the BASE.

(iii) **Psychosocial and mental health predictors** (UCLA loneliness, depressive symptoms, social connectedness). UCLA loneliness produced no detectable effect (CI: $[-0.009, 0.006]$). Depressive symptoms produced a small positive delta of +0.001 (CI: $[-0.008, 0.010]$). Social connectedness showed similarly minimal movement (CI: $[-0.008, 0.007]$). Chapter 2.3 documents bidirectional links between SOC (social isolation, depression, and functional health) in TILDA [30, 32]. The null deltas do not contradict these associations, they indicate that the relevant pathways are already captured within the functional and pain predictors present in the BASE.

(iv) **Physical and behavioural predictors** (grip strength, BMI, cigarettes per day, CAGE alcohol scale, history of blackout or fainting). Grip strength produced a positive delta of +0.004 (CI: $[-0.004, 0.012]$). Chapter 2 identifies grip strength as a primary FR marker in TILDA [24]; the near zero increment confirms that the functional limitation variables in the BASE already capture this FR related signal. BMI produced +0.002 (CI: $[-0.007, 0.012]$). Cigarettes per day trended slightly negative ($[-0.012, 0.002]$), and the CAGE alcohol scale contributed minimally ($[-0.004, 0.008]$). A history of black-out or fainting produced the highest BASE AUC in the study (0.714, CV 0.726±0.009), indicating a BASE model that is already performing strongly when this variable is withheld. Adding it back yielded no further gain (CI: $[-0.007, 0.007]$), suggesting its signal is captured through correlated cardio-vascular and health burden predictors already present in the BASE.

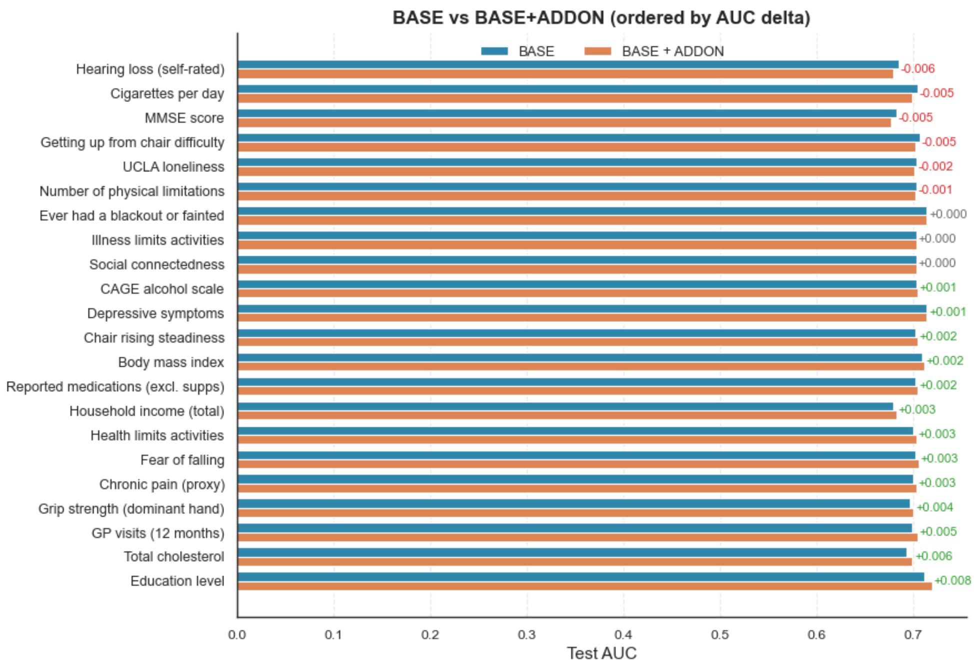

Figure 5.4: Bootstrapped 95% confidence intervals for test AUC delta (BASE+ADDON minus BASE) across all candidate features (B = 2,000 resamples). Features are sorted in ascending order by the midpoint of the delta interval. All intervals cross zero, indicating no statistically significant incremental contribution from any candidate.

Across all designs and model families, predictive signal across these related health outcomes was concentrated within a highly correlated pattern of indicators associated with accumulated health burden. Medication load, self-rated health, pain frequency and severity, and mobility difficulty collectively capture most of the discriminative signal available in this dataset: social, cognitive, nutritional, and behavioural domains showed weak positive trends but no independent contribution within the context of the available feature set and model specification. Two findings reinforce this conclusion: restricting D3 to the top-20 features caused no performance loss, and no add-on candidate improved discrimination significantly. Together, these results suggest that for 2-year transitions in this cohort, predictive signal structure reflects a compact and highly correlated set of health burden indicators rather than separable domain specific risk factors.

5.4 Health burden correlations

The results in this chapter reflect not only modelling choices but also the structure of the underlying data. This section examines pairwise correlations among the key health burden variables used in this study to explain three recurring patterns: why pooled designs outperformed the single wave baseline, why reducing the feature set caused negligible loss in performance, and why no add-on candidate produced statistically significant independent gain.

5.4.1 Shared health burden signal

When pairwise correlations among predictors approach $|r| = 1$, the effective rank of the feature matrix decreases and the marginal information gained from adding further correlated variables approaches zero, a property that directly explains why the feature reduction in D3 (Section 5.2.1.3) produced negligible performance loss, and why no add-on candidate in Subsection 5.3.4 improved discrimination significantly. Figure 5.5 presents pairwise Spearman correlations at wave 5 ($n =$ 4,978) for six health burden variables used across Track A and Track B: self-reported long term health problem, long term illness, any clinically coded chronic disease (`ICD10_ANY`), chronic pain, self-rated health, and medication use.

The most striking feature of the matrix is the perfect correlation ($r = 1.00$) between self-reported long term health problem and long term illness. Both share identical prevalence (44.2%, $n = 2{,}201$ out of 4,977 valid responses) and are effectively the same variable within this wave of TILDA. They encode a single

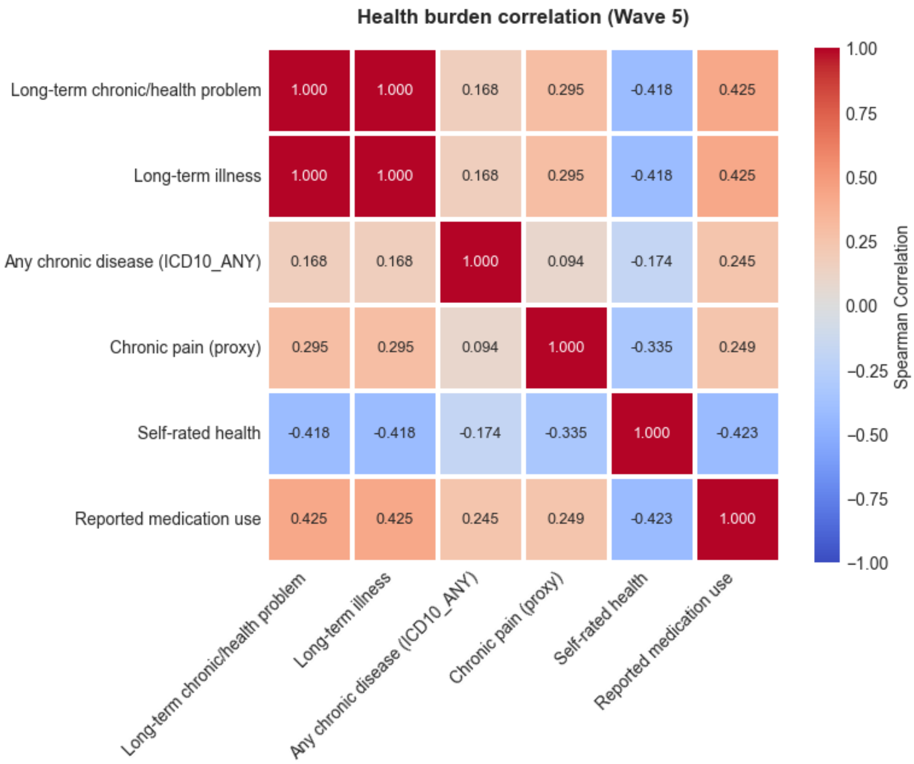

Figure 5.5: Pairwise Spearman correlations among six health burden variables at wave 5 (n = 4,978). Self-reported long term illness and chronic health problem form an identical construct (r = 1.00), while `ICD10_ANY` shows consistently weak associations with all self-reported measures.

construct, and predictive performance for either target reflects the same signal rather than two independent outcomes.

A more consequential pattern concerns the alignment between the composite ICD-10 disease indicator and the self-reported measures. Correlations range from $r = 0.094$ with chronic pain to $r = 0.245$ with medication use, with long term illness at $r = 0.168$ and self-rated health at $r = -0.174$. These weak associations are not driven by imbalance: the ICD-10 composite indicator prevalence was 37.6%, comparable to the self-reported outcomes. They reflect a genuine measurement gap, and as confirmed by the Cohen's Kappa analysis in Section 5.1.1 ($\kappa = 0.167$), the agreement between them is poor.

All top-20 variables identified in D2 and D3, including medication burden, self-rated health, chronic pain, mobility limitation, and healthcare utilisation, are survey derived and naturally aligned with the self-reported layer rather than with clinically coded records. The issue is not that these predictors are uninformative, but that they are aligned more closely with subjective and functional health states than with clinically coded disease definitions. This explains why D1, which used ICD-10 composite outcome, produced substantially weaker and less stable discrimination than D2 and D3.

A third pattern concerns the internal structure of the self-reported measures. Medication use, self-rated health, chronic pain, and long term illness are moderately correlated with pairwise values ranging from $|r| = 0.25$ to $|r| = 0.43$. Self-rated health is inversely associated with all other burden indicators: medication use ($r = -0.423$), chronic pain ($r = -0.335$), and long term illness ($r = -0.418$). These moderate correlations indicate a shared burden signal, while still leaving each measure with some independent predictive value.

This internal structure directly explains why the add-on analysis in Section 5.3 produced no statistically significant results. Add-on candidates from outside this set, including education level, grip strength, UCLA loneliness, and cognitive function (MMSE, the survey based cognitive screener available across all waves), were each tested against a BASE model already containing the core burden indicators. None produced a measurable increment, because the principal signal had already been absorbed by these variables.

Prediction of chronic disease risk in this cohort is therefore governed by a compact and internally correlated set of perceived and functional health measures. Further gains from adding variables of the same type are unlikely. The present analysis used the publicly accessible TILDA release (Waves $1-5$), which contains self-reported health, functional, cognitive, nutritional, and behavioural variables, together with ICD-10 coded disease indicators from wave 2 onwards. Richer data that could distinguish perceived health burden from objective physiological decline, including linked mortality records, primary care records, and repeated biomarker panels, are held under elevated access and were not available for this study. The inclusion of family history indicators in Track B partly reflects this constraint: participant level onset data for specific conditions were often sparse or absent in the public release, and family reported diagnoses served as an indirect proxy where direct measures were unavailable.

For a general community dwelling older adult cohort with publicly available survey data, these results suggest that chronic disease onset prediction is largely a function of how burdened a person already is, as captured through their medications, pain, and self-rated health, rather than of specific biomarker or lifestyle profiles. Extending this signal would require access to richer linked data, a direction outlined in Chapter 6.

Chapter 6

Conclusion

6.1 Summary

This thesis developed and evaluated a longitudinal ML pipeline for predicting chronic disease onset in older adults using all five waves of TILDA data. Chapter 3 described the study population and manual variable screening workflow, which reduced over 5,000 raw variables to a structured, domain-organised dataset. Chapter 4 presented the deterministic six-step cleaning pipeline and validated the transformed features against published benchmarks. Chapter 5 evaluated four model families across three modelling designs and 24 disease specific outcomes, finding that pooled multi-wave designs substantially outperformed the single-wave baseline, that predictive signal concentrated in a small cluster of health burden indicators, and that no domain specific add-on feature contributed statistically significant independent predictive value.

Code and pipeline scripts supporting all analyses are available via the repository referenced in Section 1.3.

6.2 Conclusions

Longitudinal pooling works, richer multidomain predictor sets do not. The performance and stability gains documented in Chapter 5 reflect a structural alignment between predictor horizon and outcome definition rather than simply sample size:

predicting clinically coded disease eight years out is a fundamentally harder task than predicting self-reported health burden over two-year transitions, and matching these two layers drives the gain. Restricting the feature set to 20 predictors caused no further loss. The domain specific add-on analysis confirmed that psychosocial, nutritional, lifestyle, and sociodemographic features showed no statistically significant independent contribution beyond the core health burden cluster, consistent across all four model families and all three design configurations.

Chapter 2 framed prediction as estimating $P(\text{disease} \mid X)$, where the marginal gain from any added feature depends on how much independent information it carries. The domain counts make this concrete. Of the 3,099 retained variables (Table 3.4), sociodemographic context alone contributed 844 variables, psychological and cognitive measures 291, lifestyle behaviours 120, and nutritional biomarkers 22, none of which produced a single top-20 predictor. The features that drove nearly all discriminative signal came from physical health (839 variables) and functional status (174 variables). Even then, just 20 of those 1,013 variables sufficed. Self-rated health, medication burden, and pain appear to act as integrative markers that already absorb the signal from the other domains: adding 1,277 variables across four domains that are genuinely important in ageing does not move the posterior.

The validation in Chapter 4 adds a dimension often missing from published work. Encoding decisions can materially affect model performance in health survey data. Recoding six frailty index items to respect published ordinal deficit scoring (where each item must remain bounded between 0 and 1 to preserve clinical meaning) [17] improved AUC from 0.591 to 0.636 under the biologically informed proxy, a 7.6% gain attributable entirely to encoding, with no change in the feature set or model. Health survey data carry domain specific structure that generic numeric conversion discards, and the cost can appear immediately in discrimination even when only a subset of items is affected.

The three-design structure forms a natural test of what matters. Adding temporal structure (D1 to D2) improved performance substantially. Reducing the feature set from 261 to 20 (D2 to D3) produced negligible further change, with a test AUC difference of only 0.002 for XGB, confirming that the signal is compact and concentrated. Track B reinforces the pattern: outcomes with sufficient incidence and consistent wave measurement generalised well, while outcomes depending on sparse events or indirect proxy targets did not, regardless of algorithm or feature strategy. The implication for ML on community health survey data is direct: prediction is bounded by the depth of the data, not the complexity of the model. Substantive gains beyond the current ceiling require the linked records held under elevated access (mortality registers, primary care histories, and biomarker panels), as outlined in Section 6.4.

6.2.1 Data availability

This analysis was conducted using the publicly accessible TILDA release (waves 1–5), a pseudonymised dataset covering self-reported health, physical, cognitive, nutritional, and behavioural variables, together with ICD-10 coded disease indicators from wave 2 onwards. The datasets are accessible through the ISSDA at `https://tilda.tcd.ie/data/accessing-data/`.

Several data sources relevant to chronic disease prediction are not included in the public release. These require either on-site access at TILDA's Trinity College Dublin facility or remote access through the TILDA VISTA trusted research environment. They include a dedicated biomarkers dataset, the wave 3 MRI dataset, an epigenetics dataset, linked primary care records, and linked mortality records. Access to these sources would enable future work to distinguish more precisely between perceived health burden, objective physiological decline, and verified disease trajectories.

6.2.2 Ethics statement

The studies involving human participants were reviewed and approved by the Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/participants provided their written informed consent to participate in this study.

6.3 Contributions

This research makes contributions at three levels. **Empirically**, it demonstrates that pooled longitudinal modelling substantially improves predictive performance and stability for chronic disease onset in the TILDA cohort, and identifies a compact cluster of self-reported health burden indicators as the primary driver of that signal. Domain features outside this core set showed no statistically significant independent predictive value, a result consistent across all model families and designs. **Methodologically**, a transparent, reproducible preprocessing pipeline for aligning heterogeneous longitudinal health survey data is documented in full, addressing a gap rarely covered in published ML literature. **Clinically**, the results provide a data-driven basis for risk stratification using publicly available survey data, and identify the conditions under which more complex predictor sets or model families are unlikely to improve performance without access to richer linked data.

Situating these results within the literature reviewed in Chapter 2 (Table 2.3) provides a useful reference point. The LR baseline in D2 achieved a test AUC of 0.709, comparable to the 0.78 reported for mortality prediction in TILDA [35] given that mortality is a harder, lower-prevalence target. The XGB model in D2 reached 0.715, below the 0.854 reported by [19] for diabetes specifically, though that study selected 105 specialist predictors from over 40,000 variables using a stacked ensemble with a neural network layer. D2 achieves comparable performance for a broader outcome with substantially simpler machinery, using a standard boosting model and publicly accessible features. The pooled design also improved stability well beyond what Donoghue et al. [20] reported for falls ($AUC \leq 0.69$) and what the baseline D1 achieved (0.53–0.58). Across model families in D3, the AUC range of 0.712–0.718 is both higher and substantially more stable than prior single-wave TILDA prediction work, supporting the core argument that design choices drive performance more than algorithm selection.

6.4 Future work

The recently released sixth wave of TILDA data (December 2025), together with the Intellectual Disability Supplement and the COVID-19 sub-study, would extend the longitudinal coverage and introduce health trajectory data spanning a pandemic period, likely altering self-reported burden profiles in meaningful ways. Cross-cohort validation using the RAND harmonised TILDA dataset, which maps variables to international standards compatible with cohorts such as HRS and SHARE, would test how well the identified health burden cluster generalises beyond the Irish context. Finally, access to the restricted biomarker panels, primary care records, and linked mortality data held within TILDA's secure research environment would allow future work to test whether the signal limits identified here reflect the nature of chronic disease risk itself or the constraints of the publicly available dataset.

The encoding challenges encountered in preparing the TILDA datasets point to a wider gap in the ML literature: standardised preprocessing protocols and auditable encoding indices for longitudinal health survey data do not yet exist as a shared resource. Establishing reproducible, domain-aware transformation workflows would reduce the risk that performance differences across studies reflect encoding choices rather than genuine signal, supporting more reliable cross study comparisons in health data science.

Appendix A

Script for loading/exporting PMF metadata to CSV

```
import pandas as pd
import pyreadstat
from pathlib import Path
import json

data_files = {
    "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01_TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
    "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data/0053-02_TILDA_Wave2_PMF_v2.7.sav",
    "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data/0053-04_TILDA_Wave3_PMF_v3.6.sav",
    "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05_TILDA_Wave4_PMF_v4.4.sav",
    "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06_TILDA_Wave5_PMF_v5.5.sav",
    "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data/0053-03_TILDA_Harmonized_v1.2.dta",
}


def load_df_with_var_of_interest(
    var_dir="var_of_interest", data_files=data_files, verbose=False
```

```python
):
    dfs = {}
    var_dir = Path(var_dir)

    for json_name, data_path in data_files.items():
        file = var_dir / json_name
        with open(file, "r") as f:
            vars_list = json.load(f).get("variables_of_interest")
            if verbose:
                print(f"Variables of interest from {file.name}: {vars_list[:10]} ...")

        if data_path.endswith(".sav"):
            df_full, meta = pyreadstat.read_sav(
                data_path, encoding="latin1", apply_value_formats=True
            )
            df, _ = pyreadstat.read_sav(
                data_path,
                usecols=vars_list,
                encoding="latin1",
                apply_value_formats=True,
            )
        else:
            df_full = pd.read_stata(data_path, convert_categoricals=True)
            df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]

        if verbose:
            print(
                f"Data shape (Original): {df_full.shape[0]} rows x {df_full.shape[1]} columns"
            )
            print(f"Data shape (Cleaned): {df.shape[0]} rows x {df.shape[1]} columns")
            print("-----------------------------------------------")

        dfs[file.stem] = df

```

```python
    return dfs


if __name__ == "__main__":
    for json_name, path_str in data_files.items():
        path = Path(path_str)
        wave_name = json_name.replace(".json", "")
        if path_str.endswith(".sav"):
            # SPSS
            df, meta = pyreadstat.read_sav(
                path, encoding="latin1", apply_value_formats=True
            )

        else:  # Stata .dta file
            df, meta = pyreadstat.read_dta(path, encoding="latin1")

        pd.DataFrame(
            {"variable": meta.column_names, "label": meta.column_labels}
        ).to_csv(Path("output") / f"{wave_name}_variable_descriptions.csv", index=False)

        print(f"--------------- wave {path} ---------------")
        print(df.shape)
        print(df.columns[:20])
        print(
            "Variable/Feature descriptions, csv file saved successfully -> dir=output/"
        )
        df.to_csv((Path("output") / f"{wave_name}.csv"), index=False)

    dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Cleaning pipeline scenario tables

Table B.1: Initial raw feature counts categorised by nine scenarios (pre cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	**809**	**482**	**639**	**592**	**577**	**453**

Table B.2: Scenario count progression: post step 1 (empty / blank handling)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.3: Scenario count progression: post step 2 (sentinel standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.4: Scenario count progression: post step 3 (numeric cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.5: Scenario count progression: post step 4 (censored / ordinal mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.6: Scenario count progression: post step 5 (binary classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.7: Final cleaned dataset: post step 6 (multi-class / mixed parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Appendix C

Variable scenario detection function

```python
# Sentinel values (numeric and string)
# -1 and -1.0 is sentinel but will be used in the clean pipeline
standardised_sentinels = {'-1', '-1.0', -1, -1.0} #  set -> duplicates removed -> counts correct -> single value detection is zero
numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
                     95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}" for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "refuse", 'none', 'no response', 'nan', 'not answered', 'no answer',
     'not applicable','no consent','dont know',"don't know","didn't answer",'prefer not to say', 'task not done', 'insufficient sample', 'unable to record',
     '__', '*']
blanks = ['', ' ', '\t', '\n']

SCENARIOS = [
    "Empty / blank values",
    "Numeric with sentinel",
    "Clean numeric",
    "Censored / bracketed numeric",
    "Binary categorical",
    "Multi class categorical / mixed",
    "Single value categorical",
    "Special symbol / mixed",
```

```python
    "Unknown / complex mixed"
]


# flatten all dtypes across the waves to float and category
def flatten_column_types(df):
    df_flat = df.copy()

    for col in df_flat.columns:
        if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric columns
            df_flat[col] = df_flat[col].astype(float)  # Convert to float
        else:
            df_flat[col] = (  # Convert to string, strip whitespace, lowercase, then category
                df_flat[col]
                .astype(str)
                .str.strip()
                .str.lower()
                .astype('category')
            )

    return df_flat

def detect_column_scenario_mapping(df, sample_size=10):
    summary = []

    # Regex patterns
    special_symbols_pattern = r'[^A-Za-z0-9\s\.,\-+&/()]' # special symbols excluding common numeric formatting
    bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[<>+-]"  # patterns like "10-20", "30/40", ">=50"
    sentinel_set = set(x.lower() for x in string_sentinels) # Lowercase string sentinels for comparison

    for col in df.columns:
        col_data = df[col].copy()
        dtype = str(col_data.dtype) # get data type as string

```

```python
        # Convert to string for sentinel / blank detection
        str_vals = col_data.astype(str).str.lower().str.strip() # lowercase and strip whitespace
        clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(str_vals.isin(blanks))] # remove sentinels and blanks

        # Remove sentinels and blanks for scenario detection only
        vals_no_sentinel = clean_vals[~clean_vals.astype(str).str.lower().isin(sentinel_set)
                                      & ~clean_vals.astype(str).str.lower().isin(blanks)].unique()
        num_unique_no_sentinel = len(vals_no_sentinel)

        # Check if any sentinel values are present
        sentinel_present = str_vals.isin(sentinel_set).any()

        # Detection flags
        empty_detected = str_vals.isin(blanks).any() or len(col_data) == 0

        # Numeric detection: coerce to numeric, NaNs for non-numeric
        numeric_vals = pd.to_numeric(clean_vals, errors='coerce') # coerce non-numeric to NaN
        numeric_like = numeric_vals.notna().all() and len(clean_vals) > 0 # all values are numeric-like after cleaning

        sentinel_detected = str_vals[~str_vals.isin(standardised_sentinels)].isin(sentinel_set).any() # or col_data.eq(-1.0).any() # check for sentinel presence
        censored_detected = clean_vals.astype(str).str.contains(bracketed_numeric_pattern).any() # check for bracketed numeric patterns

        # Binary categorical detection (exactly 2 unique values after cleaning)
        binary_categorical_detected = False
        if dtype == 'category':
            if num_unique_no_sentinel == 2:
```

```
                  binary_categorical_detected = True
              elif num_unique_no_sentinel == 1 and sentinel_present:
                  binary_categorical_detected = True

          # Multi-class categorical (3+ unique values, object/category)
          multi_class_categorical_detected = num_unique_no_sentinel > 2 and dtype == 'category'

          # Single value categorical
          single_value_detected = num_unique_no_sentinel == 1 and not binary_categorical_detected

          # Special symbols detection (ignore numeric formatting)
          symbol_detected = clean_vals.astype(str).str.contains(special_symbols_pattern).any()

          # Mixed types detection
          mixed_detected = len(set(type(v) for v in col_data)) > 1  # more than one data type present

        # Determine scenario in order of pipeline
          if empty_detected:
              scenario = SCENARIOS[0]
              handling = "Replace numeric/categorical empty/blank with -1.0"
          elif numeric_like and sentinel_detected:
              scenario = SCENARIOS[1]
              handling = "Replace numeric sentinel values with -1.0"
          elif numeric_like and not sentinel_detected:
              scenario = SCENARIOS[2]
              handling = "Keep as numeric"
          elif censored_detected:
              scenario = SCENARIOS[3]
              handling = "Convert numeric ranges to midpoints or keep as categorical"
          elif binary_categorical_detected:
              scenario = SCENARIOS[4]
              handling = "Encode as 0/1 or keep as category"
```

```
        elif multi_class_categorical_detected:
            scenario = SCENARIOS[5]
            handling = "One-hot encode, label encode, or keep as category"
        elif single_value_detected:
            scenario = SCENARIOS[6]
            handling = "Keep as category / constant column"
        elif symbol_detected:
            scenario = SCENARIOS[7]
            handling = "Inspect and preprocess based on context"
        else:
            scenario = SCENARIOS[8]
            handling = "Inspect manually / complex mixed types"

        sample_values = list(col_data.unique()[:sample_size]) # sample unique values

        summary.append({
            'Variable': col,
            'DataType': dtype,
            'UniqueValues': num_unique_no_sentinel,
            'Scenario': scenario,
            'SuggestedHandling': handling,
            'SampleValues': sample_values
        })

    return pd.DataFrame(summary)
```

Appendix D

Step 6: Cleaning function for multi-class and mixed variables

```python
def categorical_transform_S6(df): # Step6; Scenario -> Multi class categorical
    df_cleaned = df.copy()
    sent_set = set([str(x).lower() for x in string_sentinels]) | set([str(x).lower() for x in standardised_sentinels])
    numeric_sentinel_set = set(float(x) for x in numeric_sentinels)

    # Helper normaliser
    def _norm(tok):
        if tok is None:
            return None
        s = str(tok).strip().lower()
        if s in blanks:
            return ''
        return s

    # Words -> number map
    word_num_map = {
        'zero':0.0,'one':1.0,'two':2.0,'three':3.0,'four':4.0,'five':5.0,'six':6.0,'seven':7.0,'eight':8.0,'nine':9.0,
        'ten':10.0,'eleven':11.0,'twelve':12.0,'thirteen':13.0,'fourteen':14.0,'fifteen':15.0,'twenty':20.0,
```

```python
        'thirty':30.0,'forty':40.0,'fifty':50.0,'sixty':60.0,'seventy':70.0,'eighty':80.0,'ninety':90.0,
        'a few':2.0,'a couple':2.0,'few':2.0,'once':1.0,'twice':2.0,'several':3.0,'couple':2.0
    }
    def words_to_num(token):
        if token is None:
            return None
        token = token.strip().lower()
        if token in word_num_map:
            return word_num_map[token]
        try:
            return float(token)
        except Exception:
            return None

    # Regexes for parsing
    RE_TIME = re.compile(r'^\s*(\d{1,2}):(\d{2})\s*$')
    RE_PERCENT = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*%\s*$')
    RE_DIRECT_NUM = re.compile(r'^\s*-?\d+(?:\.\d+)?\s*$')
    RE_RANGE = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*[---/]\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_RANGE_TO = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*(?:to|TO)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_PLUS = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*\+\s*$')
    RE_GT = re.compile(r'^\s*(?:>=|>)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_LT_ANCHOR = re.compile(r'^(?:<|<=)\s*-?\d+(?:\.\d+)?')
    RE_NUM_FIND = re.compile(r'-?\d+(?:\.\d+)?')
    RE_WORD_TOK = re.compile(r"[a-z']+(?:\s[a-z']+)?")
    # RE_SURVEY_TIME = re.compile(r'^\s*(?:[a-z]+\s+)?(\d+(?:\.\d+)?)\s*(?:month|week|year)s?.*$', re.IGNORECASE)

    def norm_return(v):
        try:
            vf = float(v)
        except Exception:
            return None
        if vf in numeric_sentinel_set:
            return 'SENTINEL'
        return vf

```

```python
    def try_parse_token(x):
        """Return float, 'SENTINEL', or None for unknown token x"""
        if x is None:
            return None

        # preserve numeric pre transformed values
        if isinstance(x, (int, float)):
            if float(x) in numeric_sentinel_set:
                return 'SENTINEL'
            return float(x)

        s = _norm(x)
        # Extending sentinel detection for categorical parsing scenario
        if s == '':
            return 'SENTINEL'
        if 'not applicable' in s or 'does not have' in s or "don't receive" in s or "don\x92t receive" in s or 'unknown' in s:
            return 'SENTINEL'
        if s in sent_set or re.fullmatch(r'[^0-9a-z]+', s):
            return 'SENTINEL'
        if s in ('other', 'other specify') or s.startswith('other (') or s.endswith('(specify)'):
            return 'SENTINEL'

        s0 = s.replace(',', '').replace('$', '').replace(EUR, '').replace('#', '')

        s0 = re.sub(r'(\d+)-', r'\1 -', s0)
        s0 = re.sub(r'-(\d+)', r'- \1', s0)

        RE_PREFIX_NUM = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\.\s*.+$')

        m = RE_PREFIX_NUM.match(s0)
        if m:
            # If the format is 'NUM. anything', return the number as the category value
            return norm_return(float(m.group(1)))
```

```

        # time
        m = RE_TIME.match(s0)
        if m:
            h, mi = float(m.group(1)), float(m.group(2))
            return norm_return(h * 60.0 + mi)

        # percent
        m = RE_PERCENT.match(s0)
        if m:
            return norm_return(float(m.group(1)))

        # direct numeric
        if RE_DIRECT_NUM.match(s0):
            return norm_return(s0)

        # ranges
        m = RE_RANGE.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass
        m = RE_RANGE_TO.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # plus / greater than
        m = RE_PLUS.match(s0) or RE_GT.match(s0)
        if m:
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # less than
        if RE_LT_ANCHOR.search(s0) or 'less than' in s0:
```

```
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # or less / more
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*less', s0)
        if m:
            return norm_return(m.group(1))
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*more', s0)
        if m:
            return norm_return(m.group(1))

        # "more than X but less than Y", "over X but less than Y"
        m = re.search(r'(?:more␣than|over)\s*e?(-?\d+(?:\.\d+)?)\s*(?:but\s*)?(?:less␣than|under)\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # "between X and Y" / "between X - Y"
        m = re.search(r'between\s*e?(-?\d+(?:\.\d+)?)\s*(?:and|[---/])\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # survey time based
        time_map_weeks = {
            'one␣week': 1.0,
            'two␣weeks': 2.0,
            'a␣month␣(4␣weeks)': 4.0,
            'a␣month/4␣weeks': 4.0,
            'three␣months␣(13␣weeks)': 13.0,
            'three␣months/13␣weeks': 13.0,
```

```
            'six months (26 weeks)': 26.0,
            'six months/26 weeks': 26.0,
            'one year (12 months/52 weeks)': 52.0,
            'one year/12 months/52 weeks': 52.0
        }
        if s0 in time_map_weeks:
            return norm_return(time_map_weeks[s0])

        # If a category is '3 months or more', try to extract the number and use upper bound
        if 'months or more' in s0 and RE_NUM_FIND.search(s0):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                months = float(nums[0])
                return norm_return(months * 4.33) # Convert months to weeks (4.33 weeks per month)

        # Frequency calendar based
        if any(tok in s0 for tok in ('day','week','month','year','daily','never','almost every day','once','twice','thrice','three','several','couple')):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                val = (float(nums[0]) + float(nums[1])) / 2.0 if len(nums) >= 2 else float(nums[0])
            else:
                if 'once or twice' in s0:
                    val = 1.5
                else:
                    val = None
                    for w in RE_WORD_TOK.findall(s0):
                        wn = words_to_num(w)
                        if wn is not None:
                            val = wn
                            break
            if val is not None:
                if 'day' in s0:
                    if 'week' in s0:
                        return norm_return(val * 4.33)
                    if 'month' in s0:
                        return norm_return(val)
```

```
                if 'daily' in s0 or 'every day' in s0 or 'almost every day' in s0:
                    return norm_return(val * 30.0)
                return norm_return(val * 4.33)
            if 'week' in s0:
                return norm_return(val * 4.33)
            if 'month' in s0:
                return norm_return(val)
            if 'year' in s0:
                return norm_return(val / 12.0)
            return norm_return(val)


    likert_map = {
        # Likert / ordinals (expanded with quintile/lowest/highest detection)
        'strongly disagree': 0.0, 'disagree': 1.0, 'neither agree nor disagree': 2.0, 'neither': 2.0,
        'agree': 3.0, 'strongly agree': 4.0,
        'very poor': 0.0, 'poor': 1.0, 'fair': 2.0, 'good': 3.0, 'very good': 4.0, 'excellent': 5.0,
        'none': 0.0, 'not at all': 0.0, 'hardly ever': 0.0,
        'a little': 1.0, 'some': 2.0, 'a lot': 3.0, 'alot': 3.0, 'often': 3.0,
        'yes': 1.0, 'no': 0.0, 'sometimes': 2.0, 'rarely': 1.0, 'most or all of the time': 4.0,
        'pass': 2.0, 'fail': 1.0, '100': 100.0,
        # Problem severity mappings
        'no problem': 1.0,'minor problem': 2.0,'moderate problem': 3.0,'major problem': 4.0,
        # Concern level mappings
        'not at all concerned': 1.0,'somewhat concerned': 2.0,'fairly concerned': 3.0,'very concerned': 4.0,
        # Survey frequency mappings
        'daily / almost daily': 7.0, 'once a week or more': 6.0,'twice a month or more': 5.0,'about once a month': 4.0,
        'every few months': 3.0, 'about once or twice a year': 2.0,'less than once a year': 1.0,'never': 0.0,
        'daily': 7.0,'every day': 7.0,'almost every day': 7.0,'most or all of the time': 7.0,
```

```
            'every week': 6.0,'once a week': 6.0,'several times per week': 6.0,
            'twice a month': 5.0,'two a month': 5.0,'several times per month': 5.0,'once or twice a month': 5.0,
            'once a month': 4.0, 'once or twice a year': 2.0,'less than once a month': 1.0,
        }

        if s0 in likert_map:
            return float(likert_map[s0])

        # Quintile / ordinal like tokens: 'lowest', '2nd quintile', 'highest'
        if 'quintile' in s0 or 'lowest' in s0 or 'highest' in s0 or 'middle' in s0 or 'median' in s0:
                if 'lowest' in s0:
                    return norm_return(1.0)
                if 'highest' in s0:
                    return norm_return(5.0)
                if 'middle' in s0 or 'median' in s0:
                    return norm_return(3.0)
                mq = re.search(r'(\d+)(?:st|nd|rd|th)?\s+quintile', s0)
                if mq:
                    return norm_return(float(mq.group(1)))
                nums = RE_NUM_FIND.findall(s0)
                if nums:
                    return norm_return(nums[0])

        if 'adjacent' in s0: # Adjacent boxes: keep semantic average if present
            nums = RE_NUM_FIND.findall(s0)
            if len(nums) >= 2:
                return norm_return((float(nums[0]) + float(nums[1])) / 2.0)
            elif nums:
                return norm_return(nums[0])
            return None

        # Fallback: first number
```

```
        nums = RE_NUM_FIND.findall(s0)
        if nums:
            return norm_return(nums[0])

        return None

    def category_sort_key(u):
        us = _norm(u) or ''
        parsed = try_parse_token(us)
        if isinstance(parsed, (int, float)):
            return (0, float(parsed), us)
        if us == 'no' or us.startswith('no ') or us.startswith('not '):
            return (1, 0.0, us)
        if 'yes' in us:
            if 'both' in us or 'two' in us:
                return (1, 2.0, us)
            if 'one' in us:
                return (1, 1.0, us)
            return (1, 1.0, us)
        amount_keywords = ['none', 'not at all', 'hardly ever', 'a little', 'some', 'a lot', 'alot', 'often', 'sometimes', 'rarely']
        for i, key in enumerate(amount_keywords):
            if key in us:
                return (2, float(i), us)
        if 'adjacent boxes' in us or 'adjacent numbers' in us:
            nums = RE_NUM_FIND.findall(us)
            if len(nums) >= 2:
                a, b = float(nums[0]), float(nums[1])
                return (10, (a + b) / 2.0, us)
            elif nums:
                return (10, float(nums[0]), us)
            return (10, 0, us)
        return (99, 0, us)

    # Operate only on categorical columns
    cat_cols = df_cleaned.select_dtypes(include='category').columns.tolist()
    for col in cat_cols:
```

```
        uniques = list(dict.fromkeys(df_cleaned[col].tolist())) # Preserve original tokens with types (do NOT cast to str here)
        normalised_pairs = [(u, _norm(u)) for u in uniques] # Build normalised pairs keeping original token too
        # Keep tokens that are non sentinel and parseable (strings or numbers)
        non_sentinel = [u for u, nu in normalised_pairs if (nu and nu not in sent_set and nu not in blanks and try_parse_token(nu) != 'SENTINEL')]

        if not non_sentinel:
            continue

        # Computing parse results keyed by normalized string (but keep list of originals too)
        parse_results = { _norm(u): try_parse_token(_norm(u)) for u in non_sentinel }

        numeric_count = sum(1 for v in parse_results.values() if isinstance(v, (int, float)))
        sentinel_parsed_count = sum(1 for v in parse_results.values() if v == 'SENTINEL')
        all_parse_numeric = (numeric_count + sentinel_parsed_count == len(parse_results))

        if all_parse_numeric:
            # Map token string -> numeric (SENTINEL -> -1.0)
            code_map = { k: (float(v) if isinstance(v, (int, float)) else -1.0) for k, v in parse_results.items() }

            def map_to_numeric(x):
                # Preserve the already numeric cells (except numeric sentinels)
                if isinstance(x, (int, float)) and not pd.isna(x):
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
```

```
                    return -1.0
                xs = _norm(x)
                if not xs or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0
                return float(code_map.get(xs, -1.0))

            df_cleaned[col] = df_cleaned[col].apply(map_to_numeric).astype(float)

        else:
            first_seen_norm = []
            seen = set()
            for u in non_sentinel:
                nu = _norm(u)
                if nu not in seen:
                    first_seen_norm.append(nu)
                    seen.add(nu)

            parse_results = {nu: try_parse_token(nu) for nu in first_seen_norm}
            max_reserved_val = -1.0
            for nu, val in parse_results.items():
                if isinstance(val, (int, float)):
                    max_reserved_val = max(max_reserved_val, val) # Assuming any parsed float is a reserved ID if it came from the map
            start_i = int(max_reserved_val) + 1
            if start_i == 0: # Ensure we never start at 0 if 0 is a reserved category i.e for ('no')
                start_i = 1
            string_tokens = [nu for nu in first_seen_norm if not isinstance(parse_results.get(nu), (int, float))]
            sorted_strings = sorted(string_tokens, key=category_sort_key)

            code_map_strings = { norm_tok: i for i, norm_tok in enumerate(sorted_strings, start=start_i) }

            def map_mixed(x):
```

```
                if isinstance(x, (int, float)) and not pd.isna(x):# Preserve real numeric cell values
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
                    return -1.0
                xs = _norm(x)
                if xs == '' or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0

                parsed_val = try_parse_token(xs)
                if parsed_val == 'SENTINEL':
                    return -1.0
                if isinstance(parsed_val, (int, float)):# Captures 'no' (0.0), 'yes' (1.0), and other hardcoded likert values.
                    return float(parsed_val)
                if xs in code_map_strings:
                    return float(code_map_strings[xs])
                return -1.0

            df_cleaned[col] = df_cleaned[col].apply(map_mixed).astype(float)

    # Final sweep on numeric columns
    for c in df_cleaned.select_dtypes(include='float').columns:
        df_cleaned[c] = df_cleaned[c].fillna(-1.0)
        df_cleaned[c] = df_cleaned[c].replace(list(numeric_sentinel_set), -1.0)

    return df_cleaned
```

Appendix E

Main cleaning pipeline execution script

```
# Cleaning pipeline to all waves
dfs_cleaned = {}

# flattened dtype df
dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.items()} # flatten dtypes

# Original df
display(scenario_count_table(dfs_flat).style.set_caption("Original Table - pre cleaning")) # display original scenario count table

# Step 1: Clean empty/missing values
dfs_step1 = {wave: clean_empty_missing(df) for wave, df in dfs_flat.items()} # apply step 1 cleaning
display(scenario_count_table(dfs_step1).style.hide(axis="index").set_caption("Step 1: Empty / Blank - post cleaning"))

# Step 2: Clean sentinel values
dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.items()} # apply step 2 cleaning
display(scenario_count_table(dfs_step2).style.hide(axis="index").set_caption("Step 2: Sentinel - post cleaning"))

```

```python
# Step 3: Numerical Transformation
dfs_step3 = {wave: numerical_transform(df) for wave, df in dfs_step2.items()} # apply step 3 cleaning
display(scenario_count_table(dfs_step3).style.hide(axis="index").set_caption("Step 3: Numeric Transformation - post cleaning"))

# Step 4: Categorical Transformation for censored scenario
dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in dfs_step3.items()} # apply step 4 cleaning
display(scenario_count_table(dfs_step4).style.hide(axis="index").set_caption("Step 4: Category Transformation S4 - post cleaning"))

# Step 5: Categorical Transformation for binary values
dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in dfs_step4.items()} # apply step 5 cleaning
display(scenario_count_table(dfs_step5).style.hide(axis="index").set_caption("Step 5: Category Transformation S5 - post cleaning"))

# Step 6: Categorical Transformation for multi class/mixed values
dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in dfs_step5.items()} # apply step 6 cleaning
display(scenario_count_table(dfs_step6).style.hide(axis="index").set_caption("Step 6: Category Transformation S6 - post cleaning"))

dfs_cleaned = dfs_step6
```

Appendix F

Grouped variable audit script (Markdown output)

```python
def show_grouped_vars_for_scenarios(dfs_original, dfs_transformed, scenarios_to_show): # Function to show grouped variables per scenario across waves -> markdown format
    """
    Used to inspect the scenario variables/features that need handling for transformation/cleaning.
    A pre and post cleaning helper function to identify variables that need specific handling and to
    Inspect actual changes for a scenario from previous cleaning step to current step.
    """

    print("# Category Transformation Mapping")
    # print(f"\n## Scenarios: {scenarios_to_show}") # Markdown mapping dictionary header
    for wave in dfs_original.keys():
        print(f"\n### ==================== {wave.upper()} ====================")

        mapping_df = detect_column_scenario_mapping(dfs_transformed[wave]) # Get scenario mapping from transformed/cleaned to see overall changes for all variables
```

```python
        # mapping_df = detect_column_scenario_mapping(dfs_original[wave]) # Get scenario mapping from original to see untransformed values

        for scenario in scenarios_to_show:
            vars_in_scenario = mapping_df[mapping_df['Scenario'] == scenario]['Variable'].tolist() # Variables in scenario
            print(f"\n--- Scenario {scenario}: {len(vars_in_scenario)} ---") # Scenario header for pipeline git commit
            if not vars_in_scenario:
                print("\nNo variables found.")
                continue

            # Dictionary: frozen set(values) -> list of variables
            groups = {}

            for var in vars_in_scenario:
                unique_vals = list(dict.fromkeys(dfs_original[wave][var].dropna()))  # Preserve first seen order from ORIGINAL
                key = tuple(unique_vals)  # Use ordered tuple as key

                if key not in groups: # Initialise list if key not present
                    groups[key] = []
                groups[key].append(var) # Append variable to the group

            # Print grouped variables in markdown format friendly for Notion -> mapping dictionary
            for value_set, var_list in groups.items():
                print("\n*Variables:*","`" + ", ".join(var_list) + "`  ") # \n
                print(f"*Pre transformed values:* {list(value_set)}  ")

                # Pull transformed values from dfs_cleaned / specific step using preserved order
```

```python
                try:
                    df_before = dfs_original[wave][var_list[0]]
                    df_after = dfs_transformed[wave][var_list[0]]

                    # Build transformed list in same order as original values
                    transformed_vals = []
                    for orig_val in value_set: # Iterate through original values in order
                        mask = df_before == orig_val
                        if mask.any():
                            trans_vals = df_after[mask].unique().tolist() # Get unique transformed values
                            if trans_vals:
                                transformed_vals.append(trans_vals[0]) # Append first transformed value
                except Exception:
                    transformed_vals = []
                print(f"*Transformed numeric:* `{transformed_vals}`")


scenarios_to_show = SCENARIOS
# scenarios_to_show = ["Clean numeric"]
# scenarios_to_show = ["Censored / bracketed numeric"]
# scenarios_to_show = ["Binary categorical"]
# scenarios_to_show = ["Multi class categorical / mixed"]

show_grouped_vars_for_scenarios(dfs, dfs_cleaned, scenarios_to_show) # prev step == comparison to current step else current step transformed/cleaned for ready .md dictionary
```

Appendix G

Sample Markdown audit output for variable transformations

The grouped Markdown representation reveals three critical points:

1. Semantically identical categories are correctly grouped even under different variable names (`ph001` vs `ph009`), enabling direct comparison of representations that would otherwise appear fragmented across waves.

2. Numerical encoding is invariant to category ordering, preserving ordinal meaning across waves. Without this approach, positional encoding would assign inconsistent values, potentially distorting performance for downstream models.

3. For variables with a large and heterogeneous set of observed values (e.g., income bands, frequency ranges, or mixed numeric text responses), the grouped view provided essential visibility into how complex category spaces were being collapsed into stable numeric representations, supporting both validation and error detection at scale.

```
%-----------------------------------------------------------------
↪  ------%
%-----------------------MARKDOWN_MAPPING_DICTIONARY---------------
↪  ------%
%-----------------------------------------------------------------
↪  ------%

# Category Transformation Mapping

### =================== WAVE1 ====================

--- Scenario Empty / blank values: 0 ---

No variables found.

--- Scenario Numeric with sentinel: 0 ---

No variables found.

--- Scenario Clean numeric: 809 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Male', 'Female']
*Transformed numeric:* '[1.0, 0.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Fair', 'Good', 'Very good', 'Excellent',
↪  'Poor', "Don't know"]
*Transformed numeric:* '[2.0, 3.0, 4.0, 5.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* ['20-39', 10.0, -1.0, 15.0, 5.0, '60+',
↪  "Don't Know", 2.0, 8.0, 3.0, '40-59', 4.0, 7.0, 1.0, 6.0, 14.0,
↪  12.0, 16.0, 18.0, 13.0, 9.0, 0.0, 19.0, 11.0, 17.0]
```

```
*Transformed numeric:* '[29.5, 10.0, -1.0, 15.0, 5.0, 60.0, -1.0,
↪  2.0, 8.0, 3.0, 49.5, 4.0, 7.0, 1.0, 6.0, 14.0, 12.0, 16.0, 18.0,
↪  13.0, 9.0, 0.0, 19.0, 11.0, 17.0]'

--- Scenario Censored / bracketed numeric: 0 ---

No variables found.

--- Scenario Binary categorical: 0 ---

No variables found.

--- Scenario Multi class categorical / mixed: 0 ---

No variables found.

--- Scenario Single value categorical: 0 ---

No variables found.

--- Scenario Special symbol / mixed: 0 ---

No variables found.

--- Scenario Unknown / complex mixed: 0 ---

No variables found.

### =================== WAVE2 ===================

...

--- Scenario Clean numeric: 482 ---

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
```

```
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', '20 - 29', 'Less than
↪ 10', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 14.5, 24.5, 10.0, 30.0, -1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Very good', 'Good', 'Fair', 'Excellent',
↪ 'Poor']
*Transformed numeric:* '[4.0, 3.0, 2.0, 5.0, 1.0]'

...

### ==================== WAVE3 ====================

...

--- Scenario Clean numeric: 639 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001, ph009'
*Pre transformed values:* ['Very Good', 'Good', 'Excellent', 'Fair',
↪ 'Poor', 'DK']
*Transformed numeric:* '[4.0, 3.0, 5.0, 2.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', 'Less than 10', '20 -
↪ 29', '30+', -98.0, -99.0]
*Transformed numeric:* '[-1.0, 14.5, 10.0, 24.5, 30.0, -1.0, -1.0]'

...
```

G. SAMPLE MARKDOWN AUDIT OUTPUT FOR VARIABLE TRANSFORMATIONS

```
### ==================== WAVE4 ====================

...

--- Scenario Clean numeric: 592 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Good', 'Fair', 'Poor', 'Excellent', 'Very
↪  Good', 'RF', 'DK']
*Transformed numeric:* '[3.0, 2.0, 1.0, 5.0, 4.0, -1.0, -1.0]'

...

### ==================== WAVE5 ====================

...

--- Scenario Clean numeric: 577 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'
```

```
*Variables:* 'ph001'
*Pre transformed values:* ['Very Good', 'Fair', 'Excellent', 'Good',
↪  'Poor']
*Transformed numeric:* '[4.0, 2.0, 5.0, 3.0, 1.0]'

...
```

Appendix H

Track B cohort sample sizes and incident rates

Table H.1: Cohort sizes and incident rates for the 24 disease specific outcomes in Track B.

Disease	**Type**	**Obs.** n	**Inc.** n	**Inc. rate %**
Circulatory disease	Clinical	14539	1774	12.2
Endocrine/metabolic diseases	Clinical	4117	362	8.8
Eye diseases	Clinical	4396	444	10.1
Musculoskeletal diseases	Clinical	4269	277	6.5
Functional decline	Functional	14296	3217	22.5
Fear of falling	Functional	18114	2536	14.0
Diabetes	Family history	4307	276	6.4
High cholesterol	Family history	4312	517	12.0
Hypertension	Family history	3590	679	18.9
Heart disease	Family history	2770	598	21.6
Osteoporosis	Family history	4779	330	6.9
Alzheimer's/dementia	Family history	4827	309	6.4
Cancer	Family history	3971	532	13.4
Depression	Family history	4760	343	7.2
Pain analgesia	Medication	2152	775	36.0
High cholesterol	Medication	733	181	24.7
Antidepressant	Medication	4378	140	3.2
Antihypertensive	Medication	19845	893	4.5
Chronic pain (proxy)	Self-reported disease	15571	2709	17.4
Hearing loss	Self-reported disease	10780	1628	15.1
Urinary incontinence	Self-reported disease	16208	1361	8.4
Chronic pain	Self-reported disease	10914	1888	17.3
Angioplasty/stent	Procedure	9233	129	1.4
Hypertension	Doctor diagnosis	250	104	41.6

Appendix I

Item index and scoring of individual features

Table I.1: Scoring scheme for the 40 individual feature items.

Item	Variable	Scoring
Doctor diagnosed eye disease	ph105_5	0 = None; 1 = Yes
Doctor diagnosed hypertension	ph201_01	1 = Yes; 0 = No
Doctor diagnosed high cholesterol	ph201_08	0 = No; 1 = Yes
No conditions	ph201_14	1 = None; 0 = Otherwise
Doctor diagnosed arthritis	ph301_03	1 = Yes; 0 = No
No listed conditions	ph301_17	1 = None; 0 = Otherwise
Self-reported long-term chronic/health problem	ph003	1 = Yes; 0 = No; -1 = DK/RF
Long term illness/health problem	DISlongterm	0 = No illness; 1 = Illness; 1 = Limiting illness
Neoplasms	ICD10_02	0 = No; 1 = Yes
Diseases of the blood	ICD10_03	0 = No; 1 = Yes
Endocrine/metabolic disease	ICD10_04	0 = No; 1 = Yes
Mental and behavioural disorders	ICD10_05	0 = No; 1 = Yes
Diseases of the nervous system	ICD10_06	0 = No; 1 = Yes
Eye disease	ICD10_07	0 = No; 1 = Yes

Continued on next page

Item	Variable	Scoring
Circulatory disease	ICD10_09	0 = No; 1 = Yes
Respiratory disease	ICD10_10	0 = No; 1 = Yes
Digestive system disease	ICD10_11	0 = No; 1 = Yes
Musculoskeletal disease	ICD10_13	0 = No; 1 = Yes
Genitourinary disease	ICD10_14	0 = No; 1 = Yes
Family history of diabetes	ph726_01	0 = No; 1 = Yes
Family history of high cholesterol	ph726_02	0 = No; 1 = Yes
Family history of hypertension	ph726_03	0 = No; 1 = Yes
Family history of heart disease	ph726_04	0 = No; 1 = Yes
Family history of osteoporosis	ph726_06	0 = No; 1 = Yes
Family history of alzheimer's / dementia	ph726_07	0 = No; 1 = Yes
Family history of cancer	ph726_08_to_11	1 = Yes; 0 = No
Family history of depression	ph726_12	0 = No; 1 = Yes
Family history of other condition	ph726_95	0 = No; 1 = Yes
No family history of any disease	ph726_96	0 = No; 1 = Yes
Pain medication use	ph505	1 = Yes; 0 = No; -1 = DK/RF
Cholesterol medication	ph225	1 = Yes; 0 = No; -1 = DK/RF
Antidepressant use	MDantidepressant	0 = No; 1 = Yes
Antihypertensive use	MDantihypertensives	0 = No; 1 = Yes
Self-rated chronic pain	ph501 (proxy)	1 = Yes; 0 = No; -1 = DK
Self-rated chronic pain	CHRany_pain	0 = No; 1 = Yes
Self-rated hearing loss	ph145	0 = No; 1 = Yes; -1 = DK
Self-rated urinary incontinence	ph601	0 = No; 1 = Yes; -1 = DK/RF
Angioplasty/stent	ph208	0 = No; 1 = Yes; -1 = DK
Fear of falling	ph408	0 = No; 1 = Yes; -1 = DK
Antihypertensive medication use	ph202a	1 = Yes; 0 = No; -1 = DK

Appendix J

Top 20 feature importance for Track A designs D1 and D2

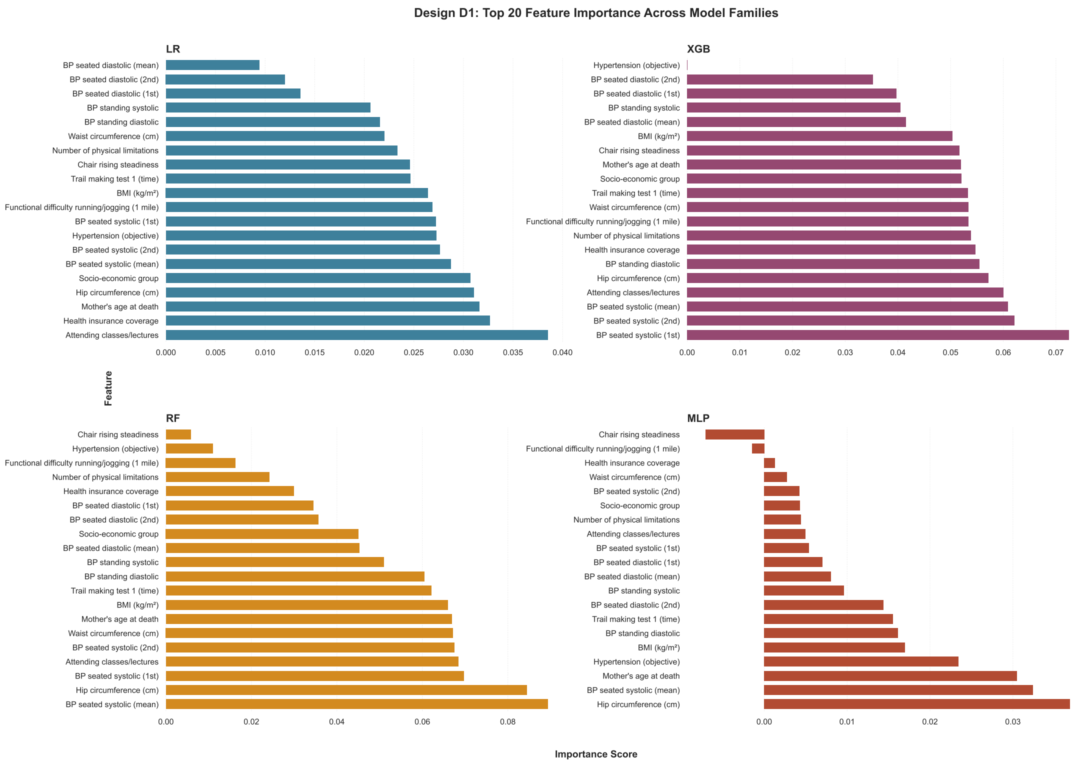

Figure J.1: Top 20 features by importance for D1.

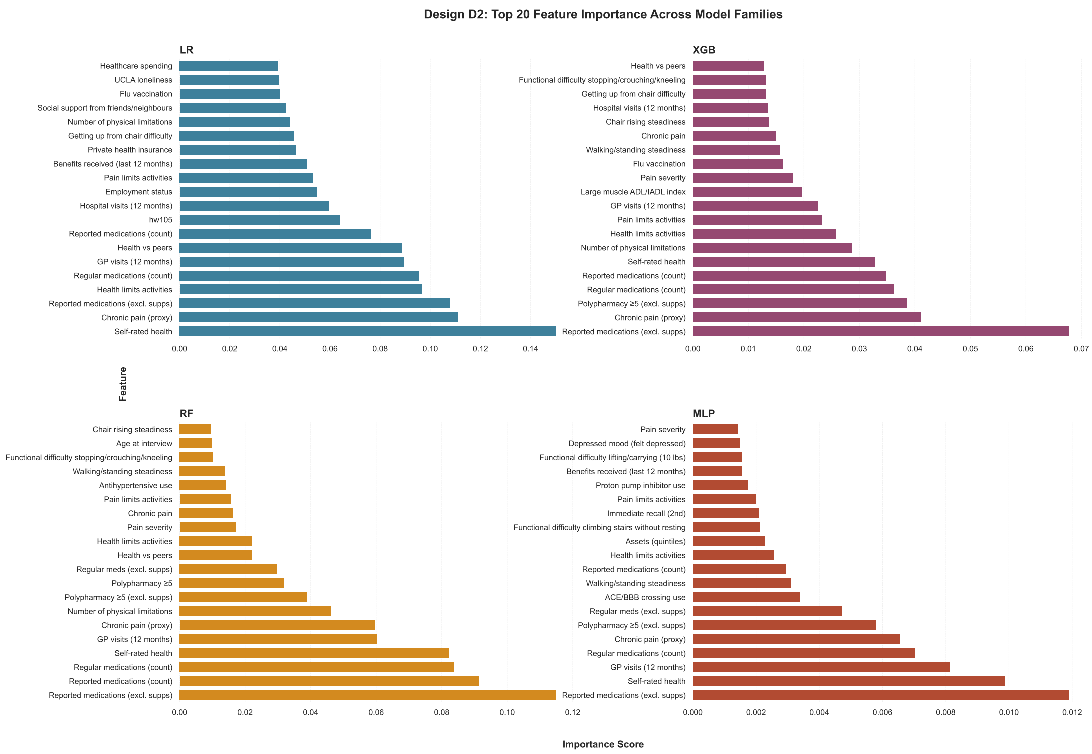

Figure J.2: Top 20 features by importance for D2.

Appendix K

SHAP value plots for Track A and Track B

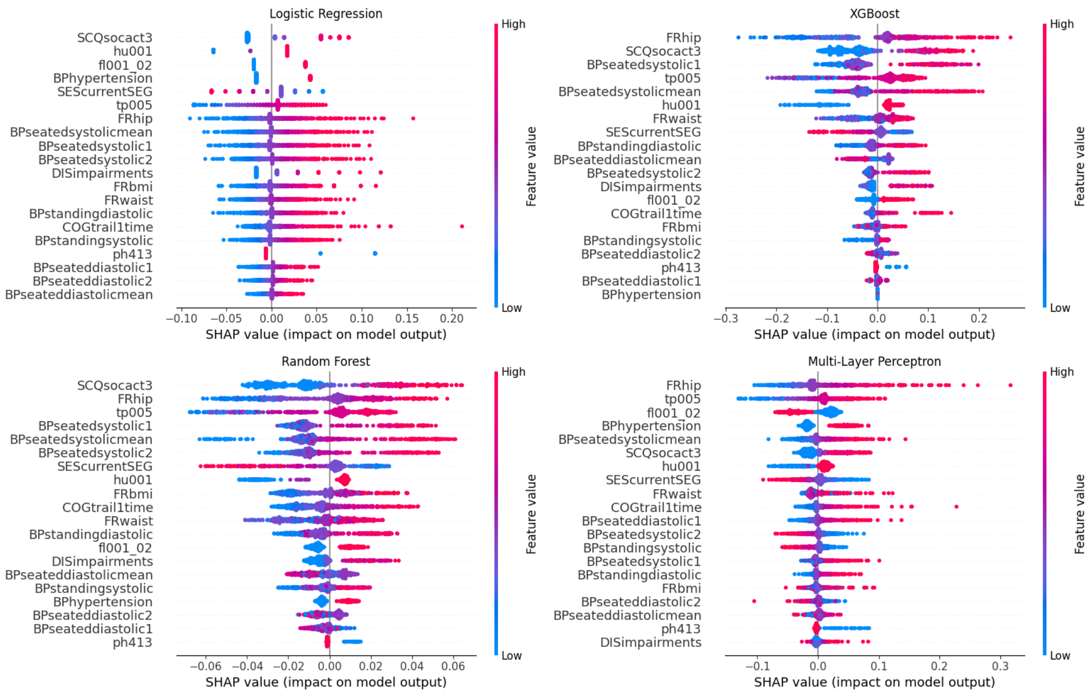

Figure K.1: Global SHAP beeswarm plots for D1 across all four model families (top-20 features by mean |SHAP|).

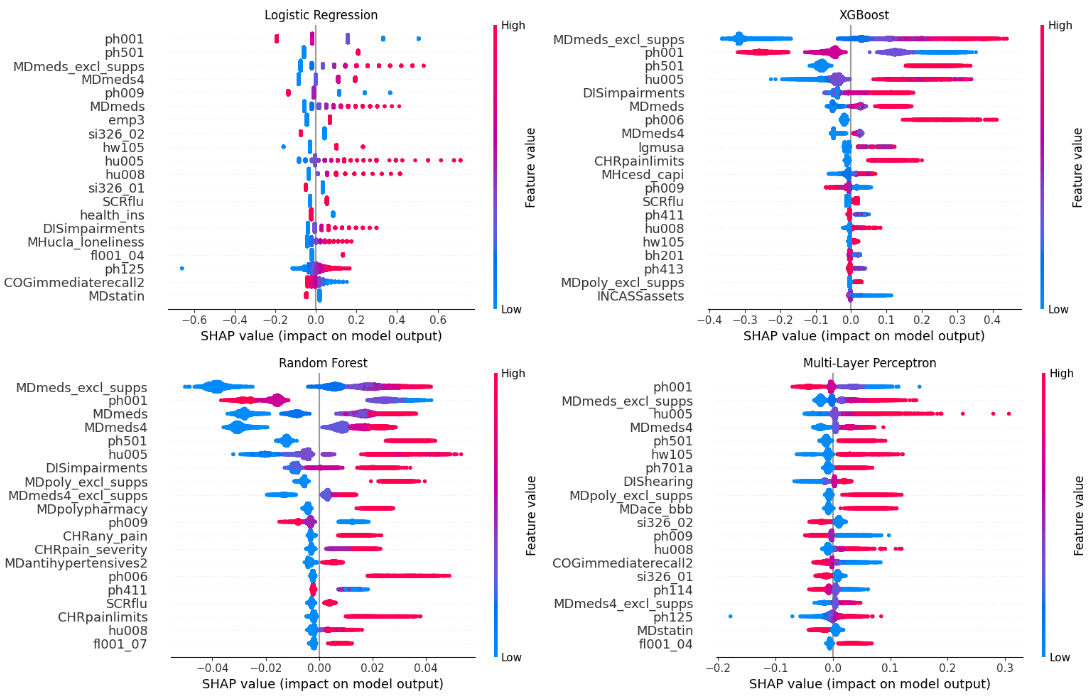

Figure K.2: Global SHAP beeswarm plots for D2 across all four model families (top-20 features by mean |SHAP|).

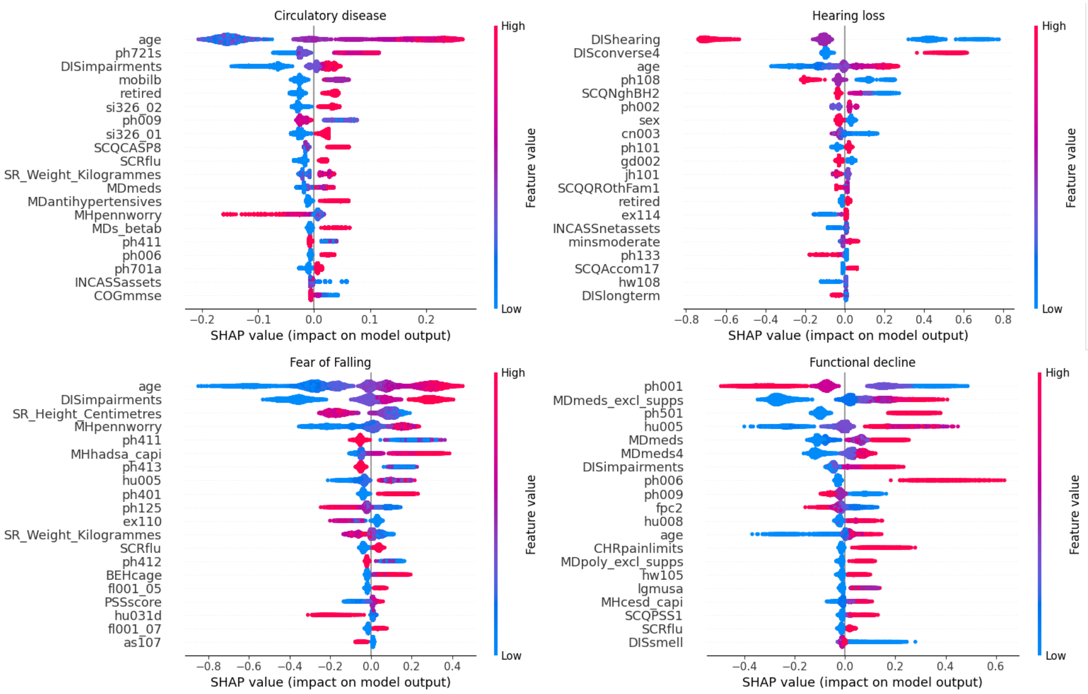

Figure K.3: Global SHAP beeswarm plots for four Track B disease specific XGB models: functional decline, fear of falling, hearing loss, and circulatory disease.

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